Permutation and Combination

Factorial Notation

$$n! = n \times (n-1) \times (n-2) \times \ldots \times 3 \times 2 \times 1$$

$$\boxed{\mathbf{n! = n \cdot (n-1)!}}$$

Fundamental Principal of Counting

5 shirts and 4 jeans →

Total No. of Possible Combinations **= 5 × 4 = 20**

5 shirts, 4 jeans, and 2 ties →

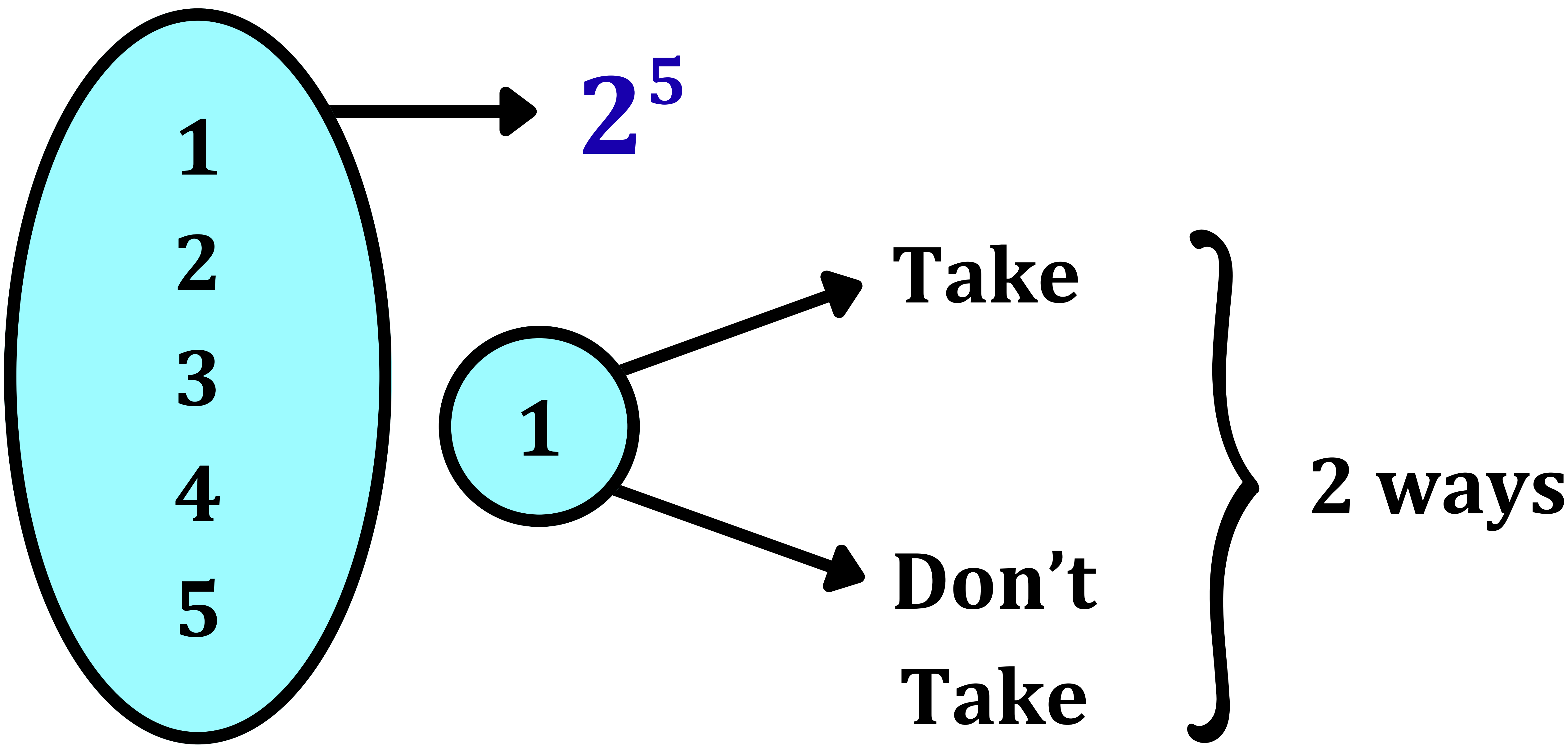
Total No. of Possible Combinations **= 5 × 4 × 2 = 40**

Definition →

If the total number of ways of occurrence of event A is "m" ways and the total number of ways of occurrence of another event B is "n" ways then →
the total number of ways of simultaneous occurrence of both the events = m · n

If the total number of ways of occurrence of event A is "m" ways, the total number of ways of occurrence of another event B is "n" ways & the total number of ways of occurrence of another event C is "p" ways then → the total number of ways of simultaneous occurrence of 3 events = $m \cdot n \cdot p$

Question → Explain how the number of Subsets in a Set containing n elements is 2^n



There are only 2 options for each element → Either they are taken into the subset or they aren't.

$$2 \cdot 2 \cdot 2 \cdot 2 \ldots (\text{n times}) = 2^n$$

Q1: The number of numbers, strictly between 5,000 and 10,000 that can be formed using the digits 1, 3, 5, 7, 9 without repetition is ____.

No. of Possibilities

$$3 \times 4 \times 3 \times 2$$

Possibilities: 5, 7 or 9

Because the numbers cannot be repeated

Answer = 3 × 4 × 3 × 2 = 72

Q2: The number of integers, greater than 7000 that can be formed using the digits 3, 5, 6, 7, 8 without repetition is ____.

Case 1 → 4 Digit Number

No. of Possibilities

$$2 \times 4 \times 3 \times 2$$

Possibilities: 7 or 8

Answer for Case 1 = 48

Case 2 → 5 Digit Number

No. of Possibilities $5 \times 4 \times 3 \times 2 \times 1$

Answer for Case 2 = 5! = 120

Answer = 120 + 48 = 168

Q3: Find the number of 5 digit integers divisible by 3 which can be formed using the digits {0, 1, 2, 3, 4, 5} without repetition.

Divisibility Rule for 3: A number is divisible by 3 if the sum of its digits is also divisible by 3.

Case 1 → The 5 Digits are 1, 2, 3, 4, 5
(1 + 2 + 3 + 4 + 5 = 15)

No. of Possibilities $5 \times 4 \times 3 \times 2 \times 1$

Answer for Case 1 = 5! = 120

Case 2 → The 5 Digits are 0, 1, 2, 4, 5

(0 + 1 + 2 + 4 + 5 = 15)

No. of Possibilities $4 \times 4 \times 3 \times 2 \times 1$

Cannot be 0

Answer for Case 2 = 4 × 4! = 96

Answer = 120 + 96 = 216

Formula for ^nP_r

$$\boxed{^nP_r = \frac{n!}{(n-r)!}}$$

Q4: If →

$^{2n+1}P_{n-2} : {}^{2n-1}P_n = 11 : 21$

then find the value of →

$n^2 + n + 15$

$$\mathbf{{}^{2n+1}P_{n-1} : {}^{2n-1}P_{n} = 11 : 21}$$

$$\mathbf{{}^{n}P_{r} = \frac{n!}{(n-r)!}}$$

$$\frac{{}^{2n+1}P_{n-1}}{{}^{2n-1}P_{n}} = \frac{11}{21}$$

$$\frac{\frac{(2n+1)!}{(2n+1-(n-1))!}}{\frac{(2n-1)!}{(2n-1-n)!}} = \frac{11}{21}$$

$$\frac{(2n+1)!}{(n+2)!} \cdot \frac{(n-1)!}{(2n-1)!} = \frac{11}{21}$$

$$\frac{(2n+1) \cdot 2n \cdot (2n-1)!}{(n+2)(n+1)n \cdot (n-1)!} \cdot \frac{(n-1)!}{(2n-1)!} = \frac{11}{21}$$

$$\frac{(2n+1)\cdot 2}{(n+2)(n+1)} = \frac{11}{21}$$

$$(2n+1)42 = 11[n^2+3n+2]$$

$$84n+42 = 11n^2+33n+22$$

$$11n^2+33n-84n-20=0$$

$$11n^2-51n-20=0$$

$$11n^2-55n+4n-20=0$$

$$11n(n-5)+4(n-5)=0$$

$$(11n+4)(n-5)=0$$

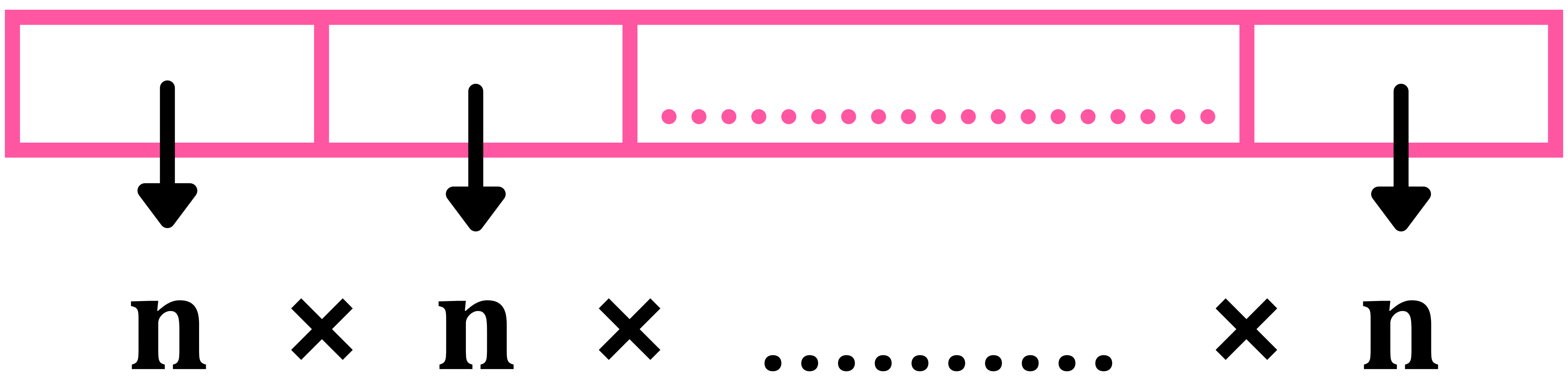
$$n=5$$

$$n^2+n+15 = 5^2+5+15 = 45$$

Theorems of Permutation (Distinct Objects)

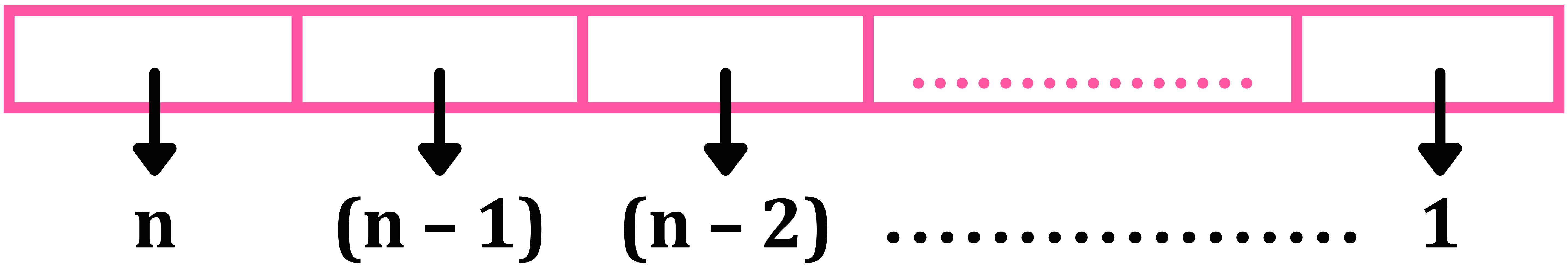
Theorem 1

The total number of permutations of n distinct objects taken all (n) at a time with repetition = n^n



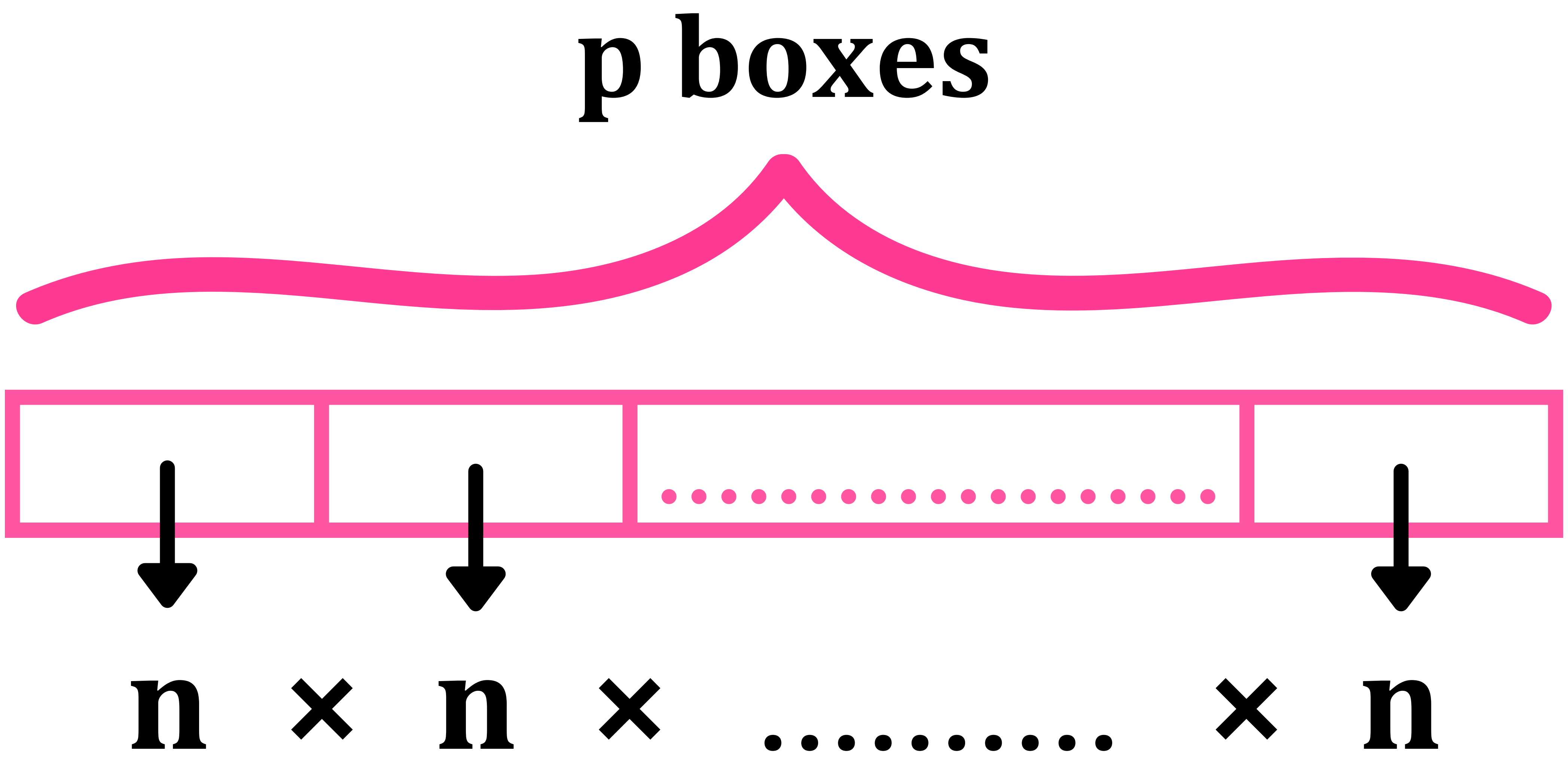
Theorem 2

The total number of permutations of n distinct objects taken all (n) at a time without repetition = n!



Theorem 3

The total number of permutations of n distinct objects taken p at a time with repetition = n^p



Theorem 4

The total number of permutations of n distinct objects taken r at a time without repetition is given by →

$$^nP_r = \frac{n!}{(n-r)!}, 1 \leq r \leq n$$

Q5: English has 26 letters. How many 5 letter combinations can be made, provided that repetition isn't allowed?

Total number of Distinct Object (26)

$$^{n}P_{r}$$

No. of objects taken at a time (5)

Answer →

$$^{26}P_{5}$$

Q6: If the digits are not allowed to repeat in any number formed by using the digits 0, 2, 4, 6, 8, then the number of all numbers greater than 10,000 is _____.

No. of Possibilities

$$4 \times 4 \times 3 \times 2 \times 1$$

Cannot be 0

Answer = 4(4!) = 96

Theorems of Permutation (Alike Objects)

Theorem 5 – Permutation of objects that are not all distinct

Say we have 3 identical A's and 4 identical B's and 6 identical C's. Then the total number of arrangements we obtain from these 13 letters is given by →

$$\underbrace{A\ A\ A}_{3!}\quad \underbrace{B\ B\ B\ B}_{4!}\quad \underbrace{C\ C\ C\ C\ C\ C}_{6!}$$

$$\text{Ans} = \frac{13!}{\underbrace{3!}_{A}\ \underbrace{4!}_{B}\ \underbrace{6!}_{C}}$$

Q7: The number of seven-digit odd numbers, that can be formed using all the seven digits 1, 2, 2, 2, 3, 3, 5 is ______.

Odd number → Ends with 1, 3 or 5.

Case 1 → 3 is the Last Digit.

					3

$$\underbrace{6 \times 5 \times 4 \times 3 \times 2}_{\text{Total no. of possible inputs}}$$

$1 \quad 2\ 2\ 2 \quad 3\ \cancel{3} \quad 5$

$3!$

$$\text{Ans} = \frac{6!}{3!} = 6 \times 5 \times 4 = 120$$

Case 2 → 5 is the Last Digit.

					5

$$1 \quad \underbrace{2\;2\;2}_{3!} \quad \underbrace{3\;3}_{2!} \quad \cancel{5}$$

$$\text{Ans} = \frac{6!}{3!\ 2!} = \frac{6 \times 5 \times 4}{2} = 60$$

Case 3 → 1 is the Last Digit.

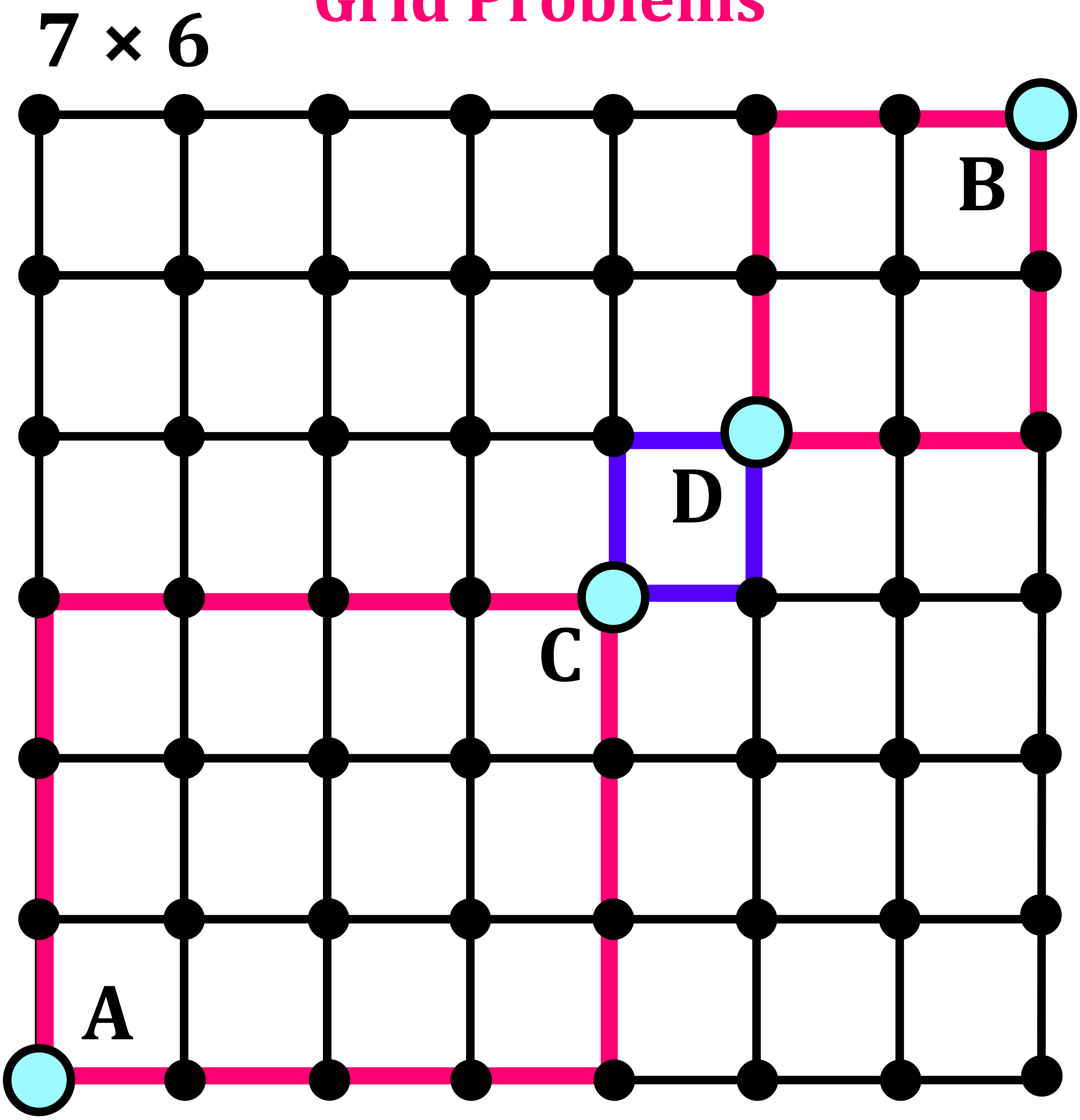
					1

$$\cancel{1} \quad \underbrace{2\;2\;2}_{3!} \quad \underbrace{3\;3}_{2!} \quad 5$$

$$\text{Ans} = \frac{6!}{3!\ 2!} = \frac{6 \times 5 \times 4}{2} = 60$$

Final Answer = 120 + 60 + 60 = 240

Grid Problems



Consider a network as shown in the figure. Paths from A to B consists of the horizontal or vertical line segments. No diagonal movement & no movement in the negative x & y direction is allowed.

1. How many paths are there from A to B?
2. How many of these paths go via C?
3. How many paths go via C and D?

There are 7 columns (Therefore, 7 X's)

X X X X X X X

There are 6 rows (Therefore, 6 Y's)

Y Y Y Y Y Y

X X X X X X X Y Y Y Y Y Y

$$\underbrace{XXXXXXX}_{7!} \quad \underbrace{YYYYYY}_{6!}$$

There are 13 total letters (7 X's + 6 Y's)

Therefore, total number of paths →

$$1.)\ \text{Ans} = \frac{13!}{7!\ 6!}$$

A is the starting point and B is the endpoint of the 7 × 6 grid. Similarly →

1. A is the starting point and C is the endpoint of the **4 × 3 grid.**
2. A is the starting point and D is the endpoint of the **5 × 4 grid.**
3. D is the starting point and B is the end point of the **2 × 2 grid.**

All paths going via point C are the same as all possible paths in the 4 × 3 grid from A to C.

No. of X's = X X X X

No. of Y's = Y Y Y

X X X X Y Y Y

$$\downarrow \qquad \downarrow$$

4! 3!

$$2.)\ \text{Ans} = \frac{7!}{4!\ 3!}$$

$$3.)\ A \longrightarrow C, C \longrightarrow D, D \longrightarrow B$$

No. of X's in the 2 × 2 grid between D & B = X X

No. of X's in the 2 × 2 grid between D & B = Y Y

X X Y Y

$$\downarrow \qquad \downarrow$$

2! 2!

$$3.)\ \text{Ans} = \frac{7!}{4!\ 3!} \times 2 \times \frac{4!}{2!\ 2!}$$

Finding Rank According to the Dictionary (Alphabetical Order)

Q8: If all the letters of the word 'VARUN' are arranged in all possible manner as they are in a dictionary, then find the rank of the word 'VARUN'

Letters in the word **'VARUN'** arranged alphabetically $\longrightarrow$

A, N, R, U, V

1st case → Words starting with A

A				
	4	3	2	1

$$\mathbf{4 \times 3 \times 2 \times 1}$$

Total no. of words starting with A $\longrightarrow$

4! = 24

(2) Words starting with N = 24
(3) Words starting with R = 24
(4) Words starting with U = 24

(5) Useful case → Words starting with V, A, N

V	**A**	**N**		

$$2 \times 1$$

Total Number of Words Starting with VAN = 2

(6) Words starting with V, A, R, N

V	**A**	**R**	**N**	**U**

The only word starting with VARN is VARNU = 1

(7) Final Case → Word is VARUN

V	**A**	**R**	**U**	**N**

Final Word = VARUN

Total Number of Words
Coming Before VARUN =
Words Starting with A **(24)**
+ Words Starting with N **(24)**
+ Words Starting with R **(24)**
+ Words Starting with U **(24)**
+ Words Starting with VAN **(2)**
+ VARNU **(1) = 99**

Therefore, the Rank of VARUN is 100.

Circular Permutation (Distinct Objects)

In a circular arrangement, all positions are initially identical. To establish a fixed reference point, one object must be fixed in place.

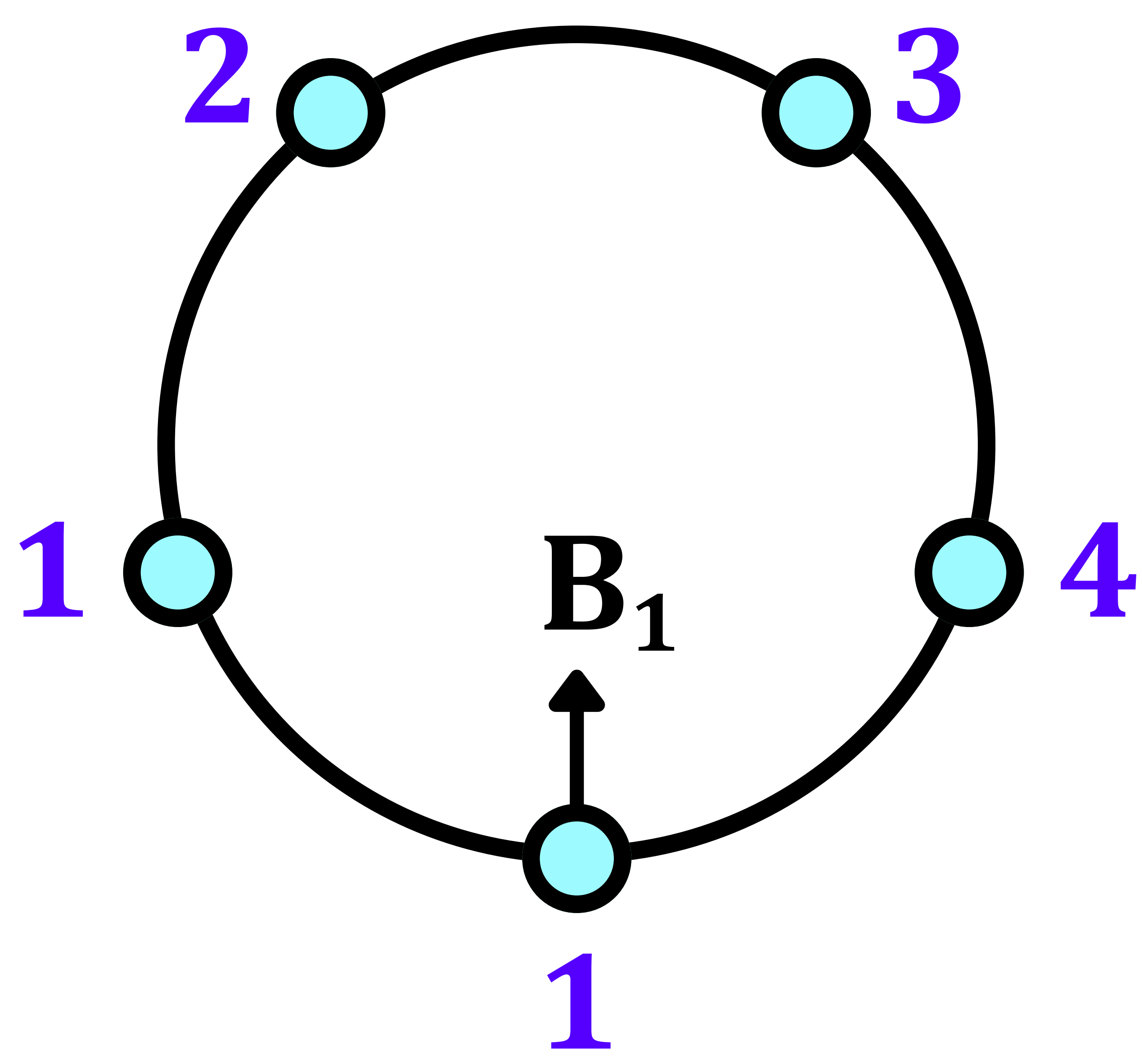
RULE 1 →

The total number of Circular Arrangements of n distinct objects is given by → (n – 1)!

Example (5 Chairs):

- Fix one boy B_1 on one chair to create a reference point.
- The remaining 4 boys can be arranged in the remaining 4 chairs in **4!** ways.

Total arrangements = 4! = 1 × 2 × 3 × 4 = 24



1 × 4 × 3 × 2 × 1 = 4! = 24

Rule 2 →

Clockwise & anticlockwise arrangements are considered the same.

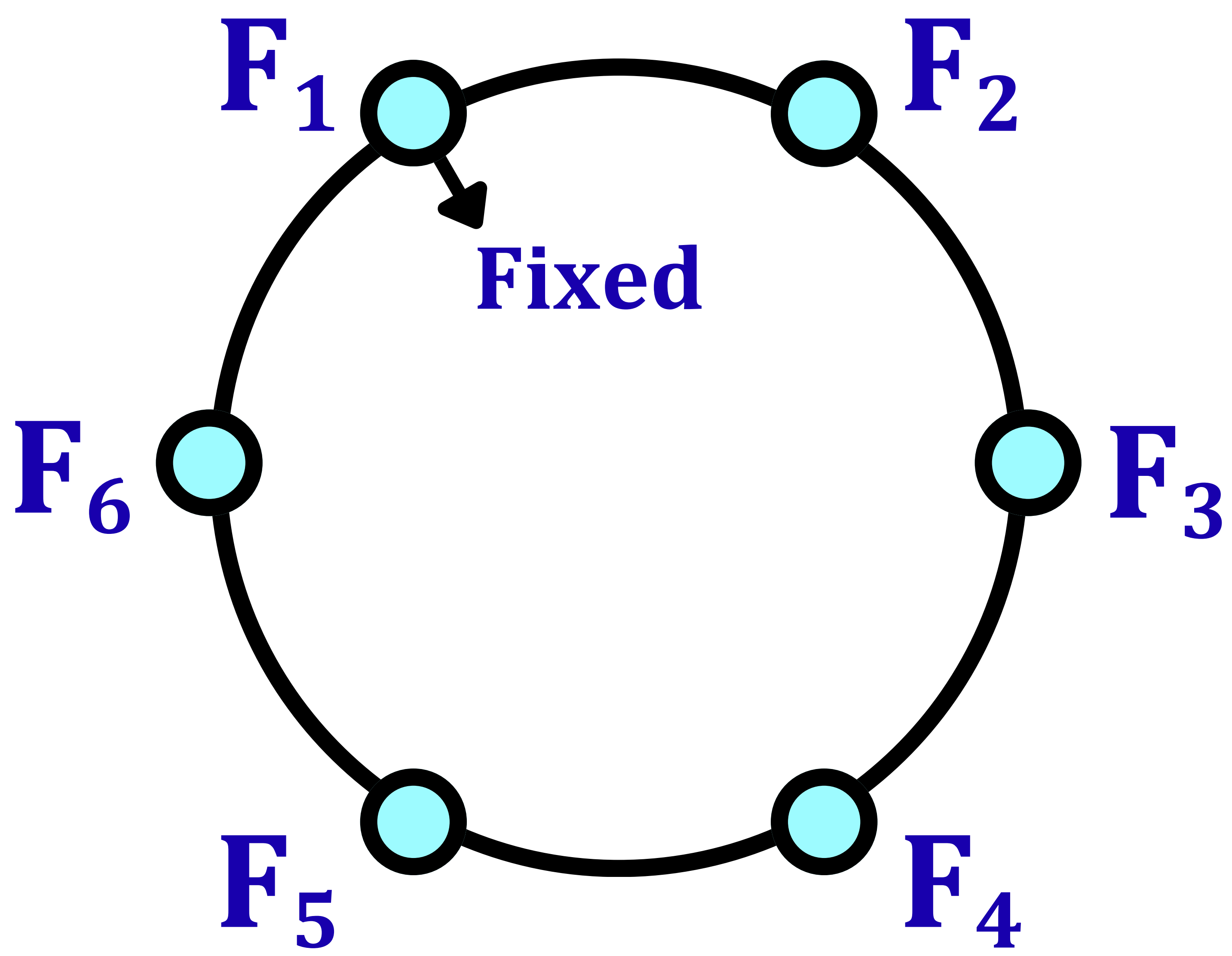
If clockwise and anticlockwise arrangements are considered identical (e.g., arrangements of beads on a necklace or flowers in a garland), the total number of unique arrangements is reduced by half.

For Example ⟶

Beads on a Necklace or Flowers on a Garland.

$$\text{Ans} = \frac{\mathbf{(n-1)!}}{\mathbf{2}}$$

Q8: How many garlands can be made using 6 different flowers?



In this case, both anticlockwise and clockwise arrangements will be considered the same.

$$\text{Ans} = \frac{(n-1)!}{2} = \frac{(6-1)!}{2} = \frac{(5)!}{2} = 60$$

Combinations → Selections (Distinct Objects)

The core difference between permutations and combinations is whether the order of selection matters.

Permutation (Order Matters)

- **Definition:** An arrangement of objects where a specific sequence or order is important. Changing the order creates a new outcome.
- **Example:** Picking 2 letters from {A, B, C, D, E} where order matters:
    - AB and BA are counted as two different outcomes.
- **Formula:** The number of permutations of r items from a set of n is given by ⟶

$$\boxed{{}^{n}P_{r} = \frac{n!}{(n-r)!}}$$

Combination (Order Doesn't Matter)

- **Definition:** A selection of objects where the order of selection is irrelevant; only the final group matters.
- **Example:** Choosing 2 letters from {A, B, C, D, E} where order doesn't matter:
  - AB and BA are considered the same single outcome.
- **Formula:** The number of combinations of r items from a set of n is given by $\longrightarrow$

$$\boxed{{}^{n}C_{r} = \frac{n!}{r!(n-r)!}}$$

$$\boxed{{}^{n}C_{r} = \frac{{}^{n}P_{r}}{r!}}$$

Q9: Assuming Cricket World Cup-2023 is played Between 10 teams. How many total matches are played in league stage if each team plays with every other team exactly once.

$$^{n}C_{r} = \frac{n!}{r!(n-r)!}$$

Here, n = 10 (total teams) and r = 2r (teams per match).

$$^{10}C_{2} = \frac{10!}{2!(10-2)!} = 45$$

The total number of matches played is 45.

Properties of ^nC_r

$$^nC_0 = {}^nC_n = 1$$

$$^nC_1 = n$$

$$^nC_r = {}^nC_{n-r}$$

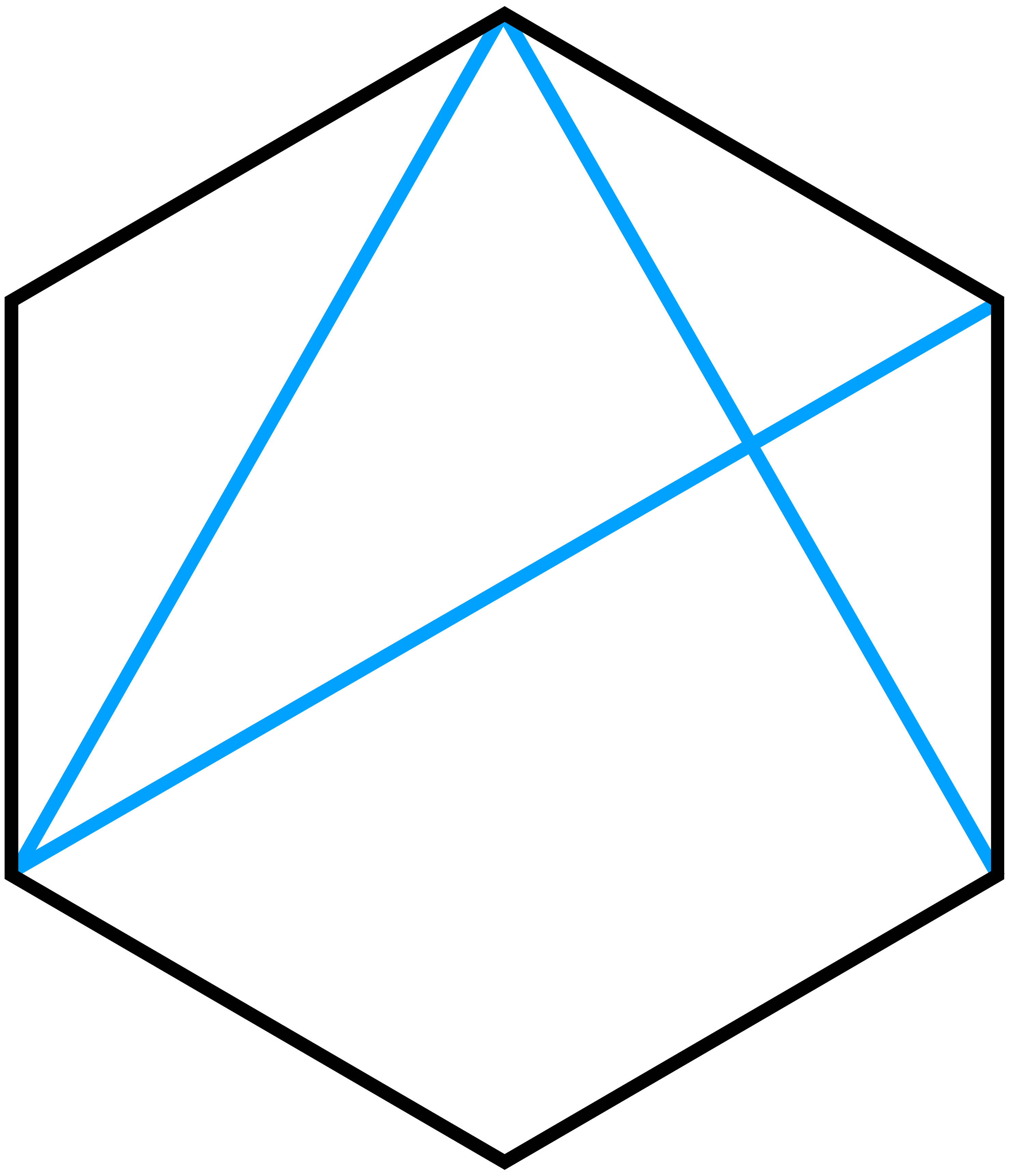
$$^nC_2 = n\frac{(n-1)}{2}$$

$$^nP_r = {}^nC_r \cdot r!$$

$$^nC_r + {}^nC_{r+1} = {}^{n+1}C_{r+1}$$

Using Combinations for Diagonals

A diagonal connects two non-adjacent vertices of a polygon.



no. of diagonals + no. of sides = $^6C_2$

no. of diagonals = $^6C_2$ – no. of sides

In general, the number of diagonals a polygon with *n* number of sides has is given by →

$$\boxed{^nC_2 - n}$$

$$^{n}C_2 - n$$

$$= \frac{n(n-1)}{2} - n$$

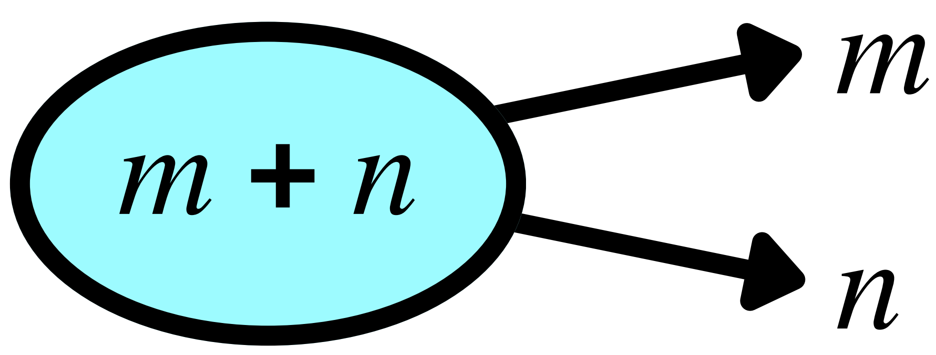
$$= \frac{n^2 - n - 2n}{2}$$

$$= \frac{n^2 - 3n}{2}$$

$$\boxed{^{n}C_2 - n = \frac{n\,(n-3)}{2}}$$

Formation of Groups

Case – 1 → Unequal Groups ($m \neq n$)



If you have $m + n$ distinct items to divide into two groups of m and n items respectively, the number of ways is a simple combination or selection.

Number of ways to divide (or select) →

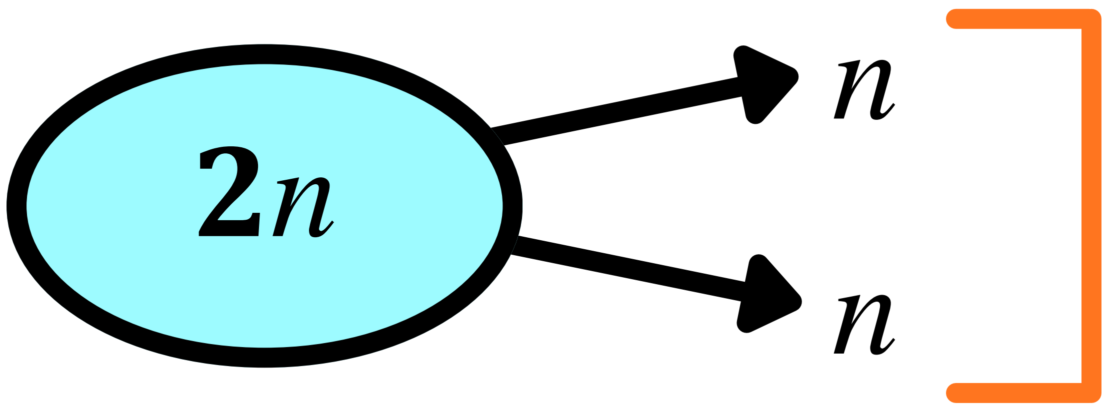
$$\frac{(m+n)!}{m!\,n!} = {}^{m+n}C_m$$

Number of ways to distribute between 2 distinct recipients →

$$\frac{(m+n)!}{m!\,n!} \times 2!$$

Because the two groups can be assigned to the two recipients in 2! ways.

Case – 2 → Equal Groups ($m = n$)



2 identical group → Divide by 2!

When the group sizes are equal, the initial division formula overcounts the number of unique groups because the two groups are indistinguishable. You must divide by 2! to correct for this.

Number of ways to divide (or select) →

$$\frac{(2n)!}{n!\,n! \times 2!}$$

Number of ways to distribute between 2 distinct recipients →

$$\frac{(2n)!}{n!\,n! \times 2!} \times 2!$$

EXAMPLE →

Let us use the example of 5 items → (A, B, C, D, E) which are to be divided into a group of 3 and a group of 2.

(m = 3, n = 2, $m \neq n$)

Division / Selection →

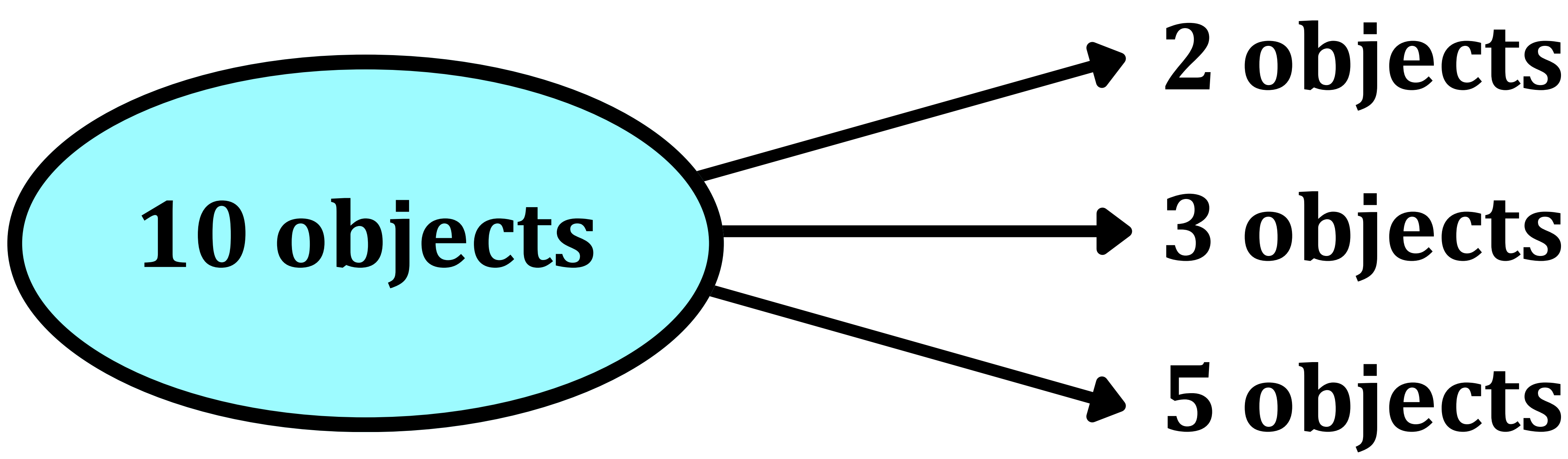
$$\frac{(m+n)!}{m!\,n!} = \frac{5!}{3!\,2!} = 10$$

Distribution Between 2 Distinct Recipients →

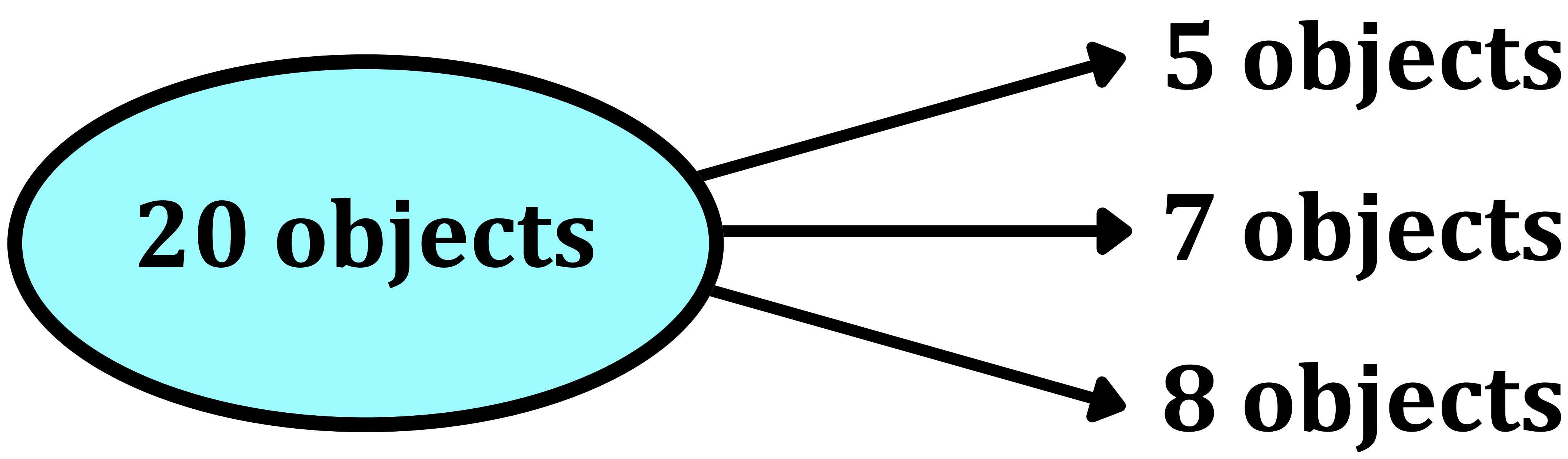
$$\frac{(m+n)!}{m!\,n!} = \frac{5!}{3!\,2!} \times 2! = 20$$

This second step accounts for the "arrangement" or "distribution" aspect where, for example, group {ABC} going to person 1 and {DE} to person 2 is different from {DE} to person 1 and {ABC} to person 2.

Division into Unequal Groups (Case 3)

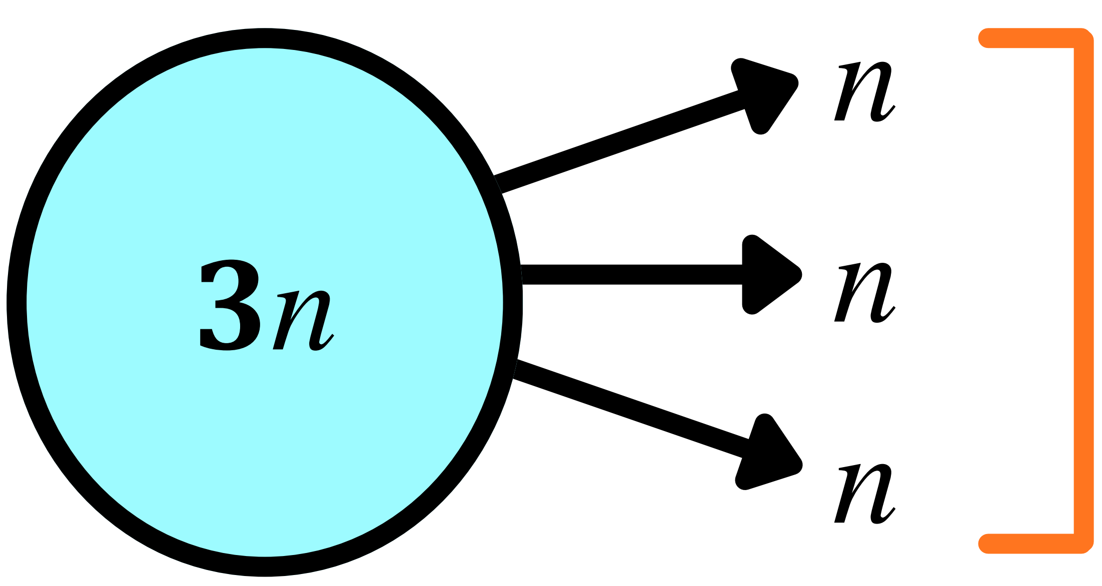


$$\frac{10!}{2!\ 3!\ 5!}$$



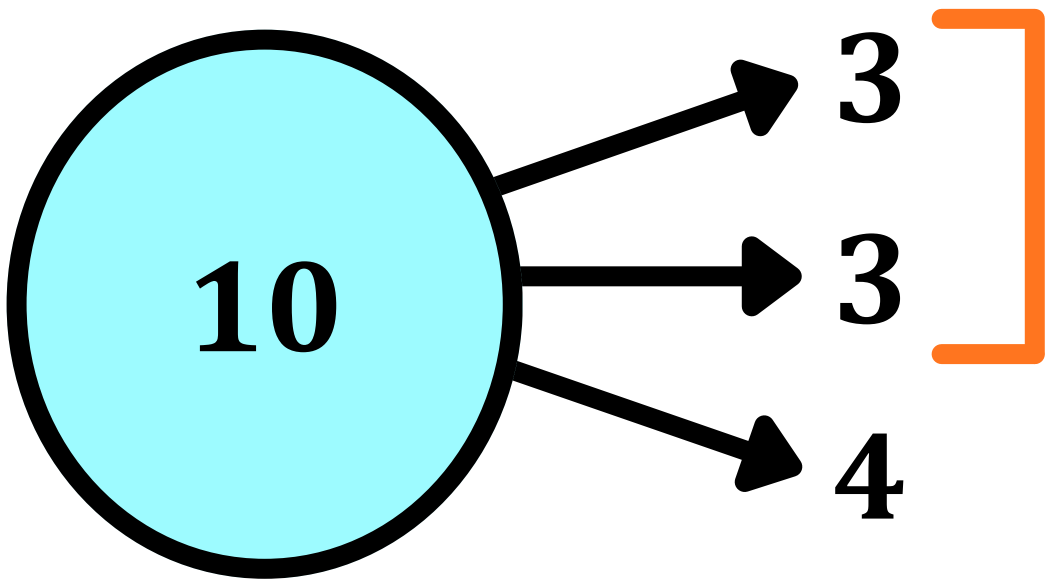
$$\frac{20!}{5!\ 7!\ 8!}$$

Division into Equal Groups (Case 5):



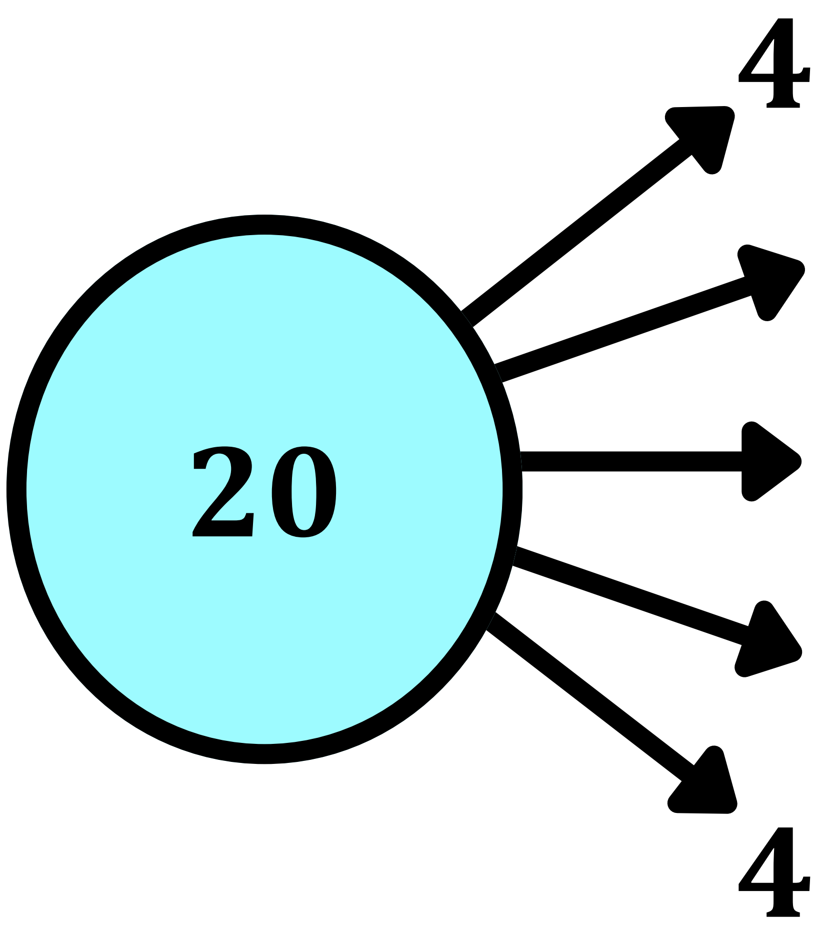
3 identical group → Divide by 3!

$$\frac{3n!}{n!\ n!\ n!\ 3!}$$



2 identical groups → Divide by 2!

$$\frac{10!}{3!\ 3!\ 4!\ 2!}$$

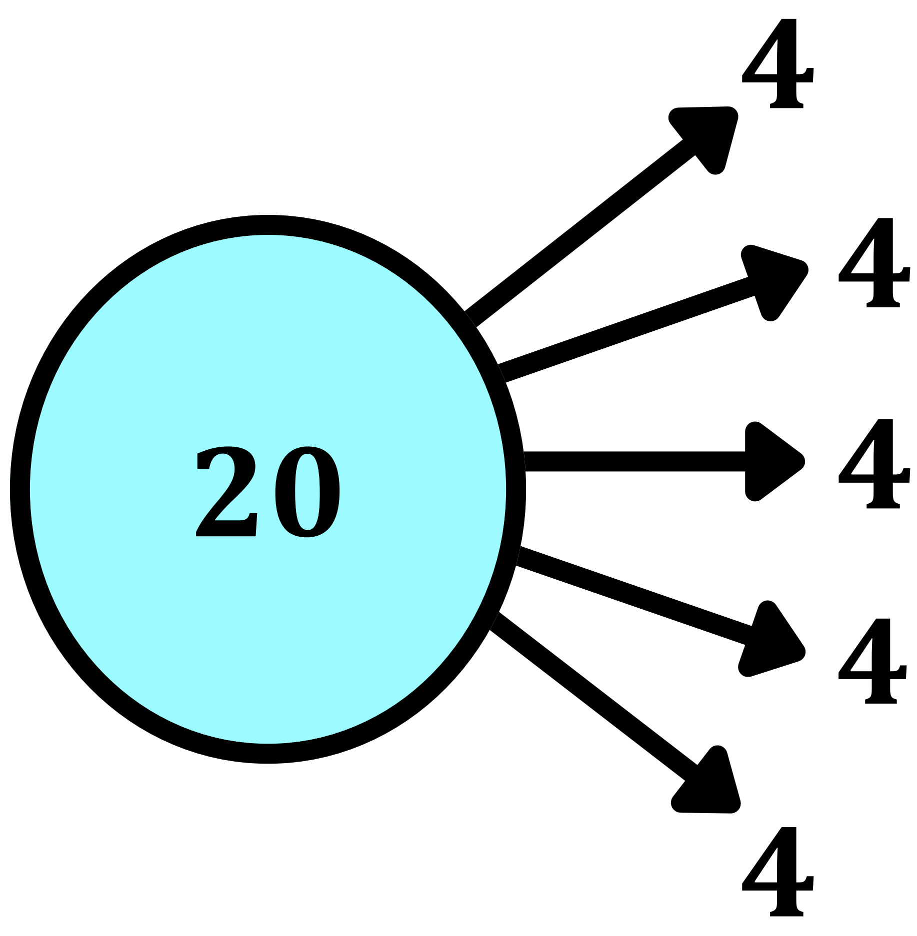


5 identical groups → Divide by 5!

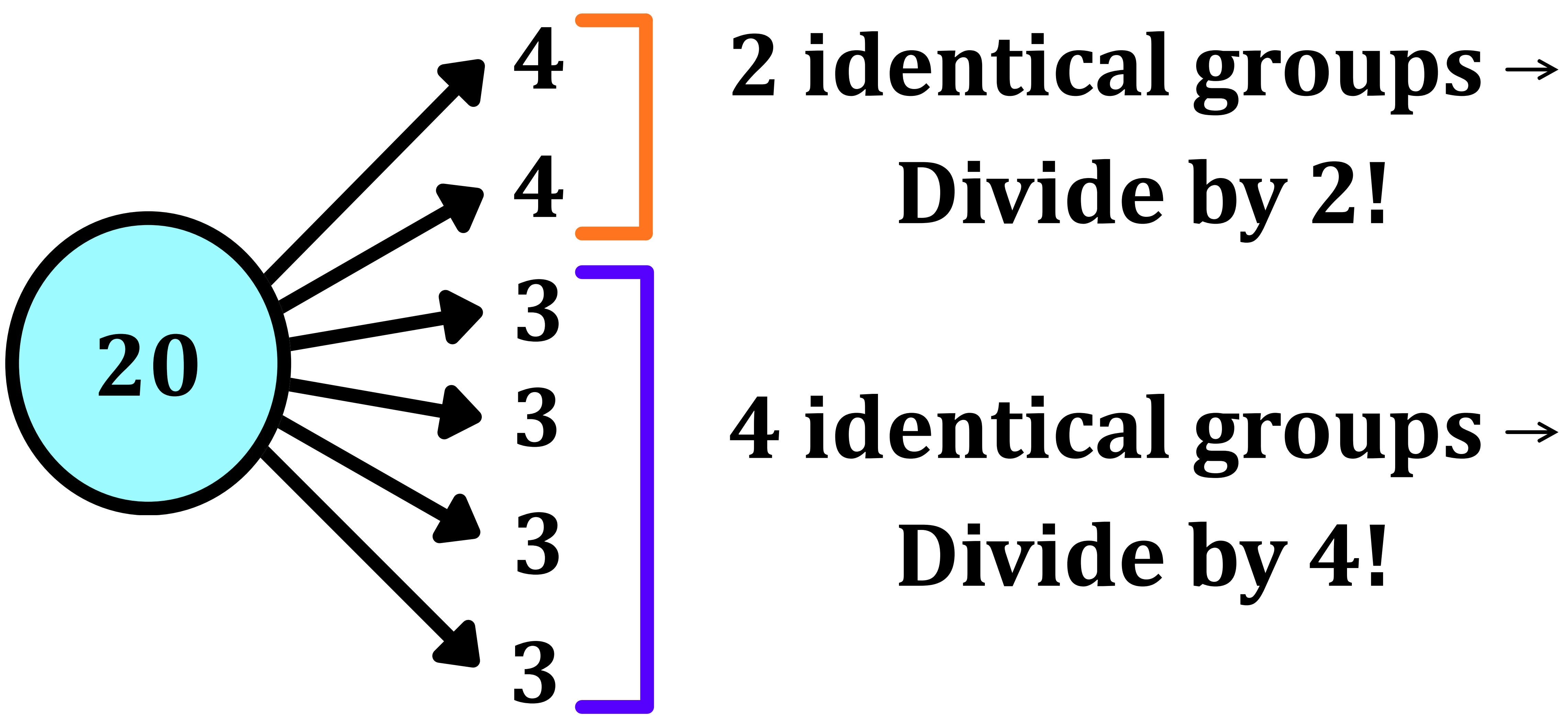
$$Division = \frac{20!}{(4!)^5\ 5!}$$

$$Distribution = \frac{20!}{(4!)^5\ 5!} \times 5!$$

For distribution, simply multiply the division with the factorial of the number of the total groups (5! in this case).

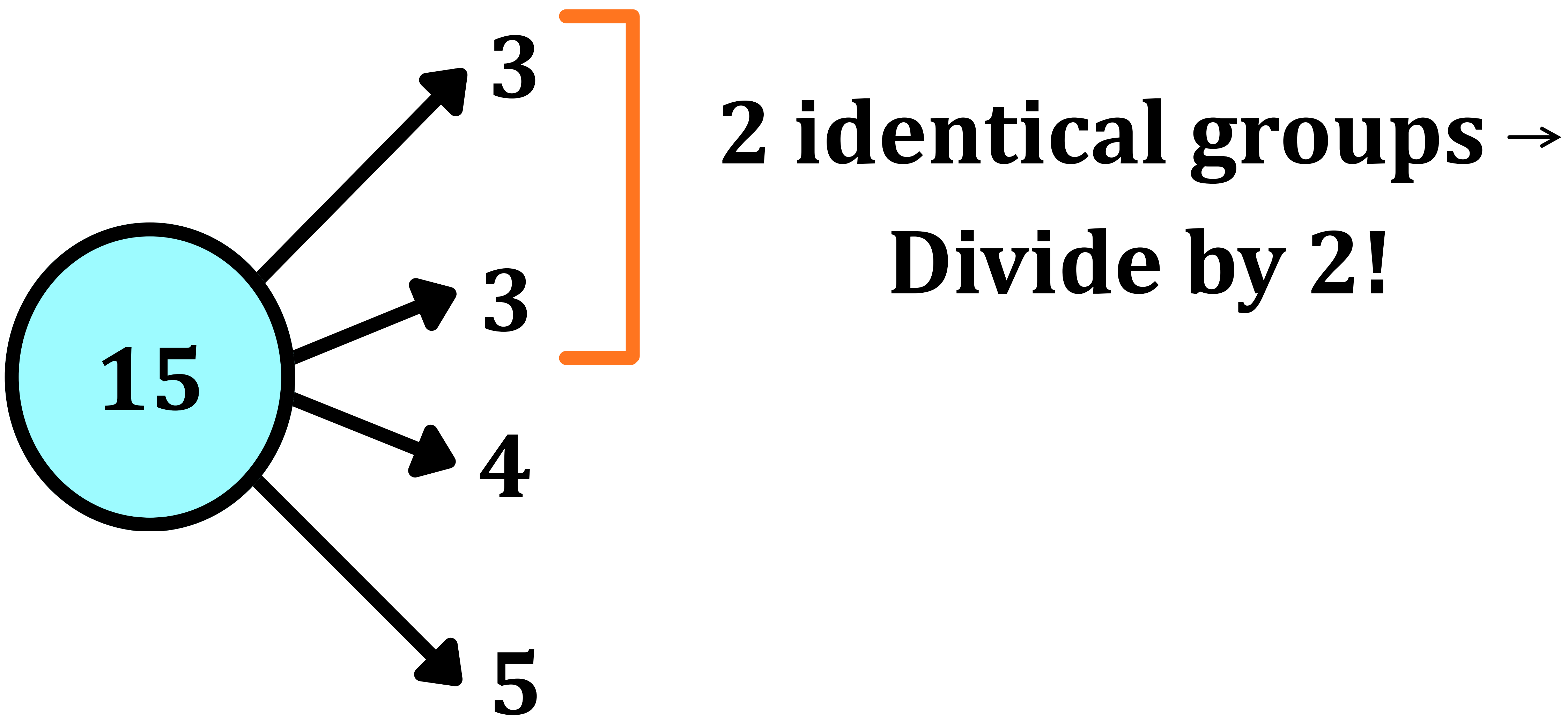


Assigning these five groups to 5 distinct people → the extra division by 5! is canceled out because the order/assignment of the groups now matters. Therefore, for distribution, we multiply this formula with 5!



$$Division = \frac{20!}{(4!)^2\ (3!)^4\ 2!\ 4!}$$

$$Distribution = \frac{20!}{(4!)^2\ (3!)^4\ 2!\ 4!} \times 6!$$



$$Division = \frac{15!}{3!\,3!\,4!\,5!\,2!}$$

$$Distribution = \frac{15!}{3!\,3!\,4!\,5!\,2!} \times 4!$$

Q10: In how many ways can 6 distinct balls be distributed to 4 boxes such that no box is empty?

Each box must at least have 1 ball.

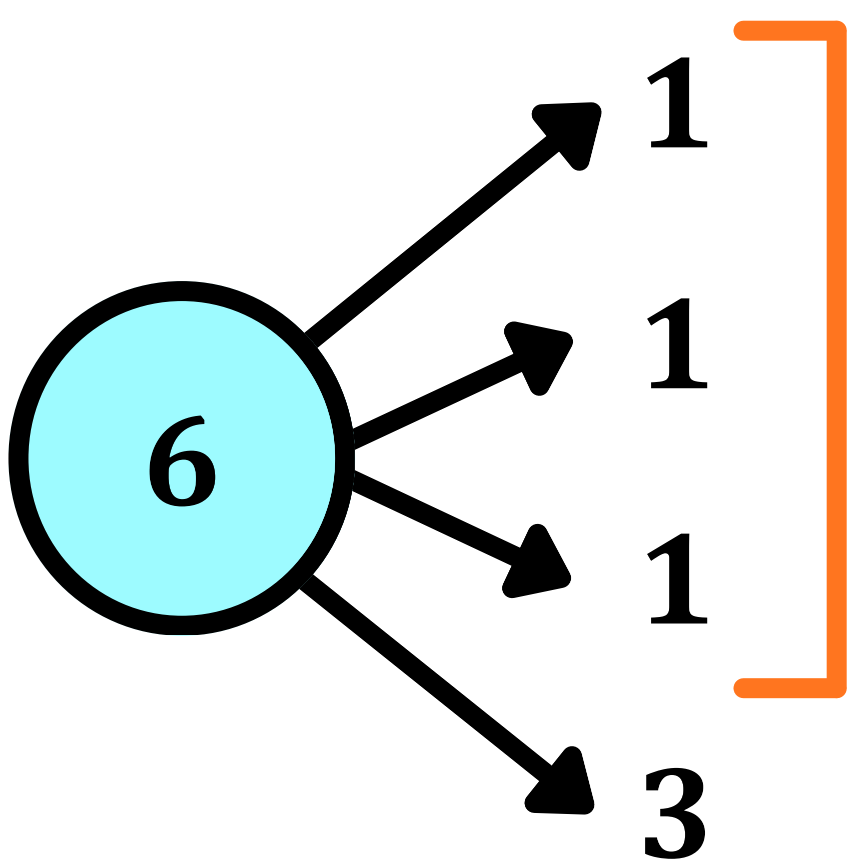
Let the 4 boxes be named B_1 B_2 B_3 & B_4

There are only 2 possible different combinations →

	B_1	B_2	B_3	B_4
Case 1	1	1	1	3
Case 2	1	1	2	2

But the order (i.e., assignment or distribution) can differ.

CASE 1 →



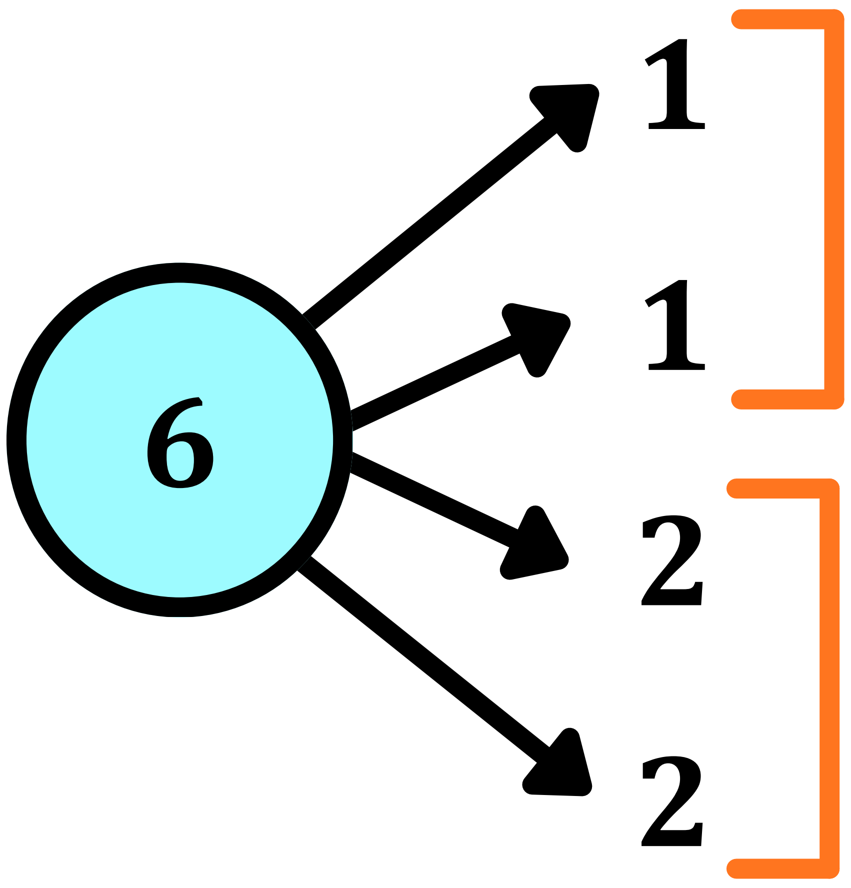
3 identical groups → Divide by 3!

$$Division = \frac{6!}{1!\,1!\,1!\,3!\,3!}$$

$$Distribution = \frac{6!}{1!\,1!\,1!\,3!\,3!} \times 4!$$

$$Distribution = \frac{6!}{3!\,3!} \times 4! = 480$$

CASE 2 →



2 identical groups → Divide by 2!

2 identical groups → Divide by 2!

$$Division = \frac{6!}{1!\,1!\,2!\,2!} \times \frac{1}{2!\,2!}$$

$$Distribution = \frac{6!}{2!\,2!\,2!\,2!} \times 4!$$

$$Distribution = 1080$$

$$Answer = 480 + 1080 = 1560$$

Selection of At Least One Object

CASE 1 → From *n* distinct objects

Let there be T_1 T_2 T_3 T_4 ... T_n objects.

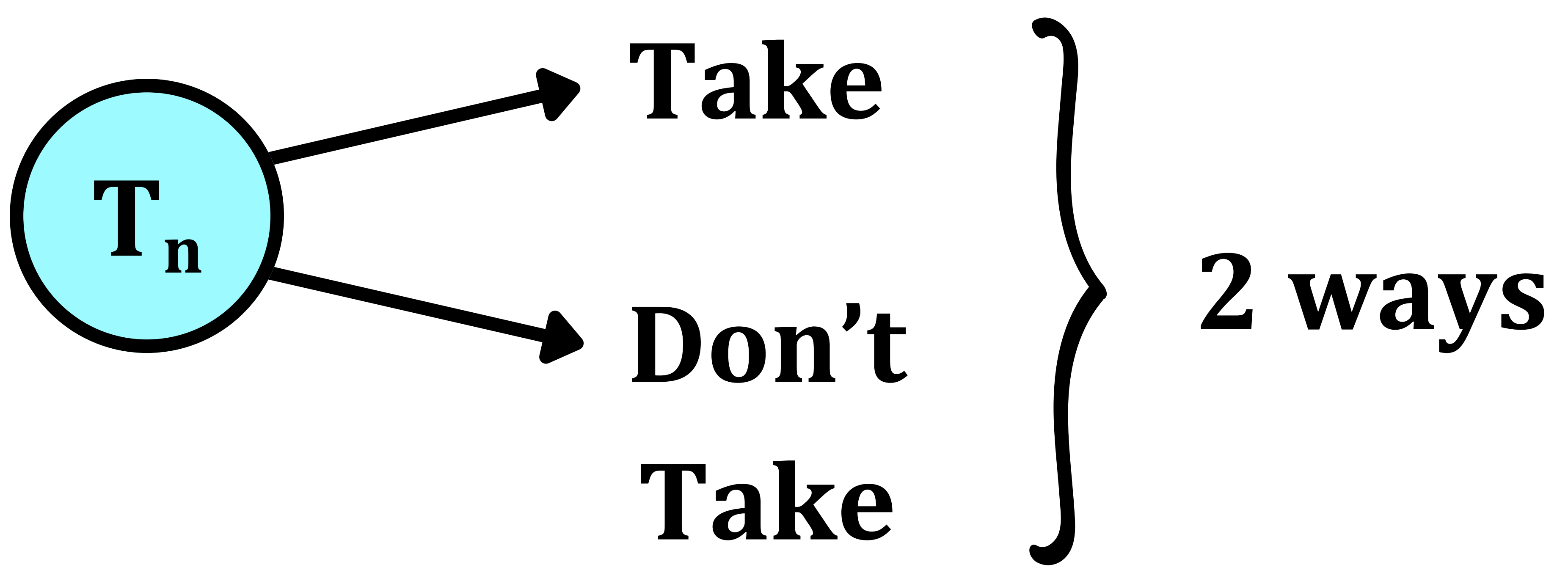
METHOD 1 →

We can only select one or more objects. Therefore, the formula for the total number of ways to choose at least one object from n distinct objects is given by →

$$^{n}C_1 + {}^{n}C_2 + {}^{n}C_3 + \cdots + {}^{n}C_n$$

METHOD 2 →

There are only two ways of dealing with each object in T_1 T_2 T_3 ... T_n and that is to either choose them or to not choose them.



Since there are 2 ways to deal with each object and there are *n* objects in total → we can say that the total number of ways to deal with these *n* objects is equal to 2^n

But there is one case where no object will be selected (i.e., "Don't take" will be the option chosen for all *n* objects).
This case needs to be subtracted because we must select AT LEAST 1 object.

Therefore, in METHOD 2, we subtract 1 from 2^n to satisfy the given condition of selecting at least one object from *n* distinct objects.

$$2^n - 1$$

EQUATING METHOD 1 AND METHOD 2 →

$$^n\mathrm{C}_1 + {}^n\mathrm{C}_2 + {}^n\mathrm{C}_3 + \cdots + {}^n\mathrm{C}_n$$
$$= 2^n - 1$$

Adding $^n C_0$ to both sides → $^n C_0 = 1$

$$^n\mathrm{C}_0 + {}^n\mathrm{C}_1 + {}^n\mathrm{C}_2 + \cdots + {}^n\mathrm{C}_n$$
$$= 2^n - 1 + {}^n\mathrm{C}_0 = 2^n - 1 + 1 = 2^n$$

$$^{n}C_0 + {}^{n}C_1 + {}^{n}C_2 + \cdots + {}^{n}C_n = 2^n$$

Q11: From a library containing 5 different physics books, 4 different chemistry books & 6 different maths books. Find the number of ways in which we can select:

1. At least 1 book.
2. At least 1 book of Each Subject.
3. At least 2 books.
4. At least 2 books of Each Subject.

Let the 5 Physics Books be P_1 P_2 P_3 P_4 P_5

Let the 4 Chemistry Books be C_1 C_2 C_3 C_4

Let the 6 Math Books be M_1 M_2 M_3 M_4 M_5 M_6

1. At least 1 book out of 15 books →

$$2^{15} - 1$$

2. At least 1 book of Each Subject →

For Physics (5 Books) → $2^5 - 1$

For Chemistry (4 Books) → $2^4 - 1$

For Math (6 Books) → $2^6 - 1$

$$(2^5 - 1) \times (2^4 - 1) \times (2^6 - 1)$$

3. At least 2 books →

Using Formula from Method 1 →

$$^{n}\mathrm{C}_1 + {}^{n}\mathrm{C}_2 + {}^{n}\mathrm{C}_3 + \cdots + {}^{n}\mathrm{C}_n$$
$$= 2^n - 1$$

$$^{n}\mathrm{C}_1 = n$$

$$^{n}\mathrm{C}_2 + {}^{n}\mathrm{C}_3 + \cdots + {}^{n}\mathrm{C}_n$$
$$= 2^n - 1 - {}^{n}\mathrm{C}_1 = 2^n - 1 - n$$

$$n = 15$$

$$2^n - 1 - n = 2^n - 16 = Ans$$

4. At least 2 books of Each Subject →

$$(2^5 - 1 - {}^{5}\mathrm{C}_1)(2^4 - 1 - {}^{4}\mathrm{C}_1)(2^6 - 1 - {}^{6}\mathrm{C}_1)$$

$$= (2^5 - 6)(2^4 - 5)(2^6 - 7)$$

CASE 2 → From *n* similar objects

Let the objects be T, T, T ... n times

METHOD 1 →

1 + 1 + 1 + 1 + ... (n times) = n

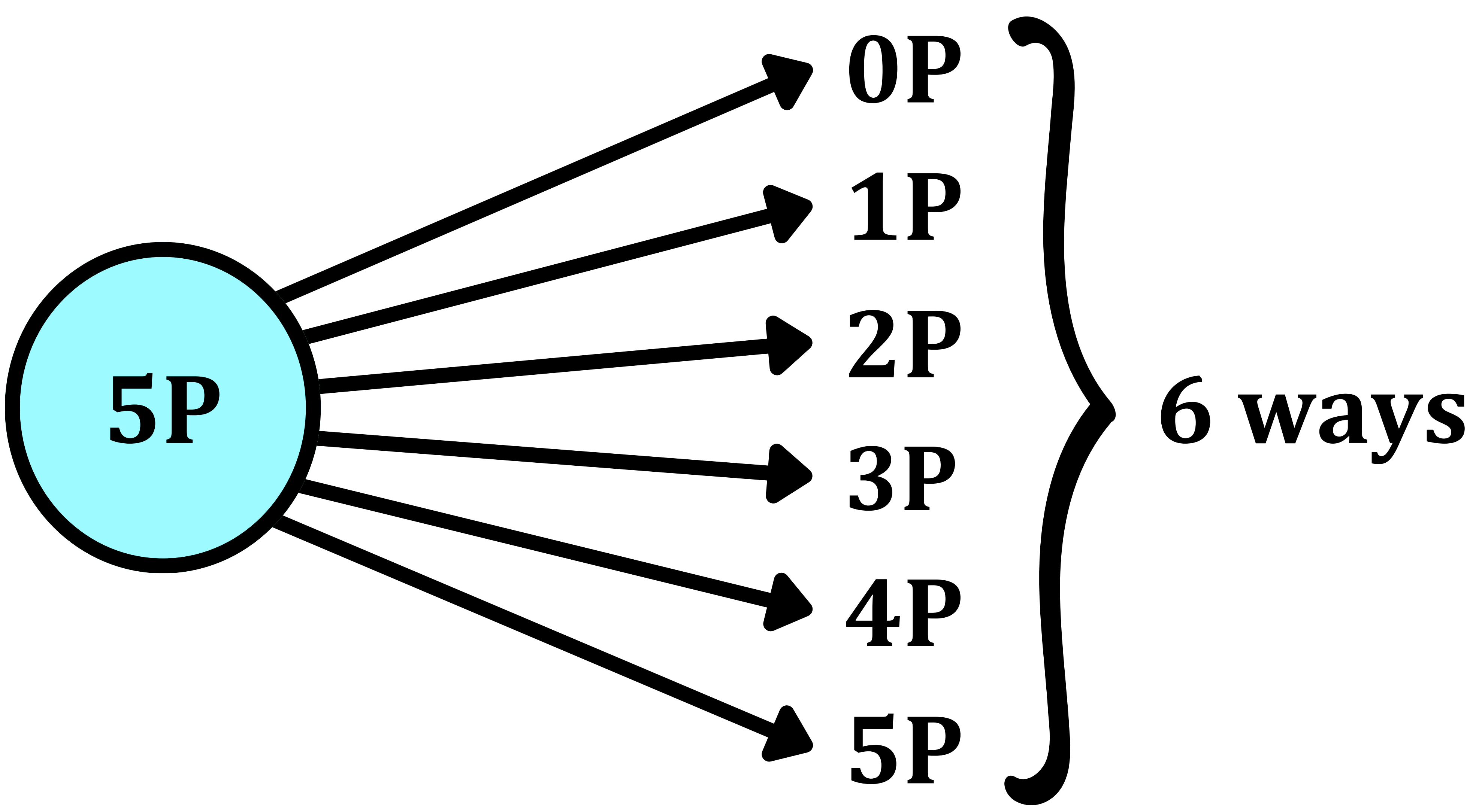
Therefore, there are *n* ways of dealing with *n* identical objects provided we choose at least one object.

METHOD 2 →

The total number of ways to deal with *n* identical objects = n + 1

This includes the possibility of not choosing any of them.

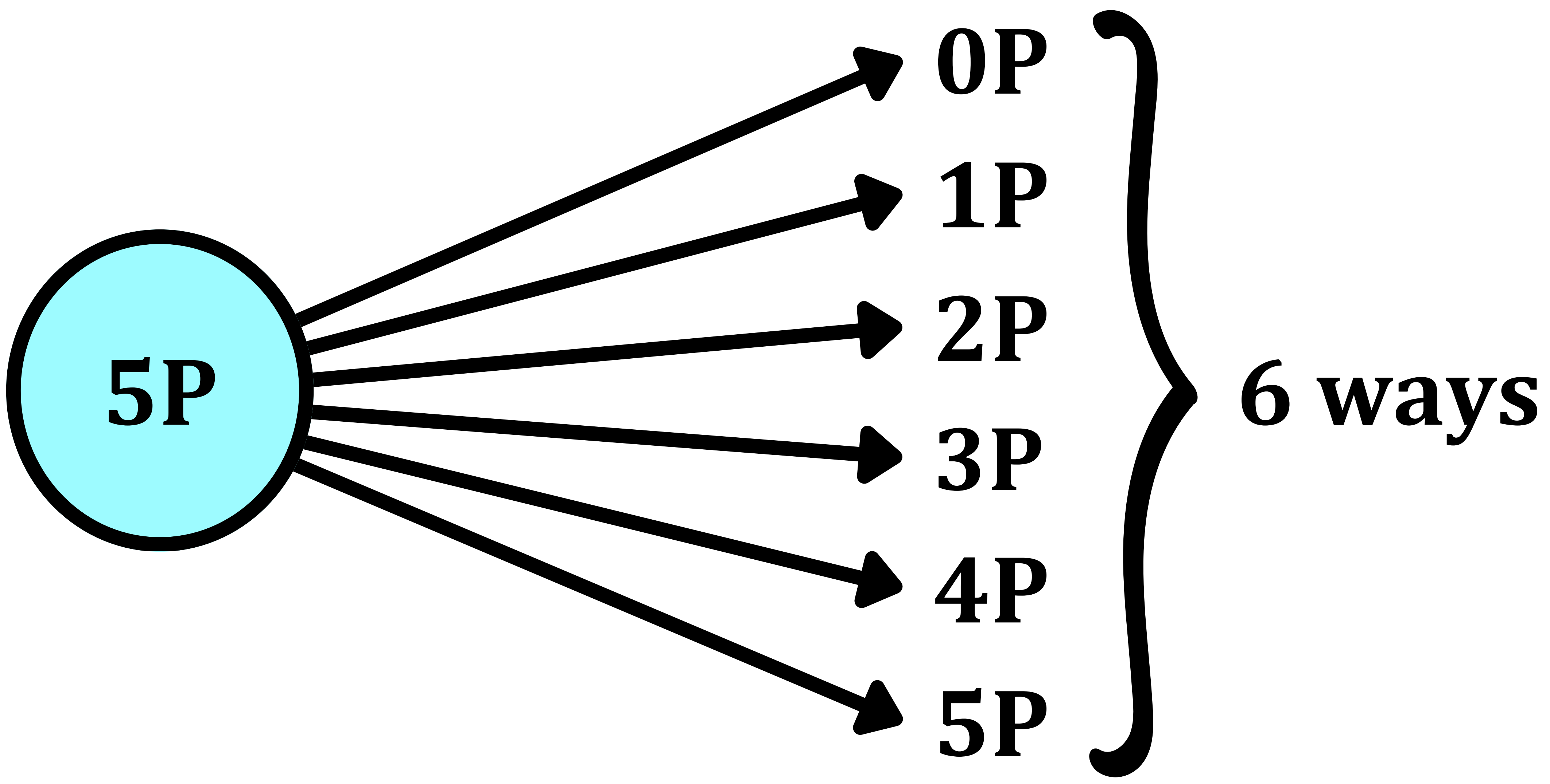
E.g., we have 5 identical pens. There are 5 + 1 = 6 ways of dealing with them.

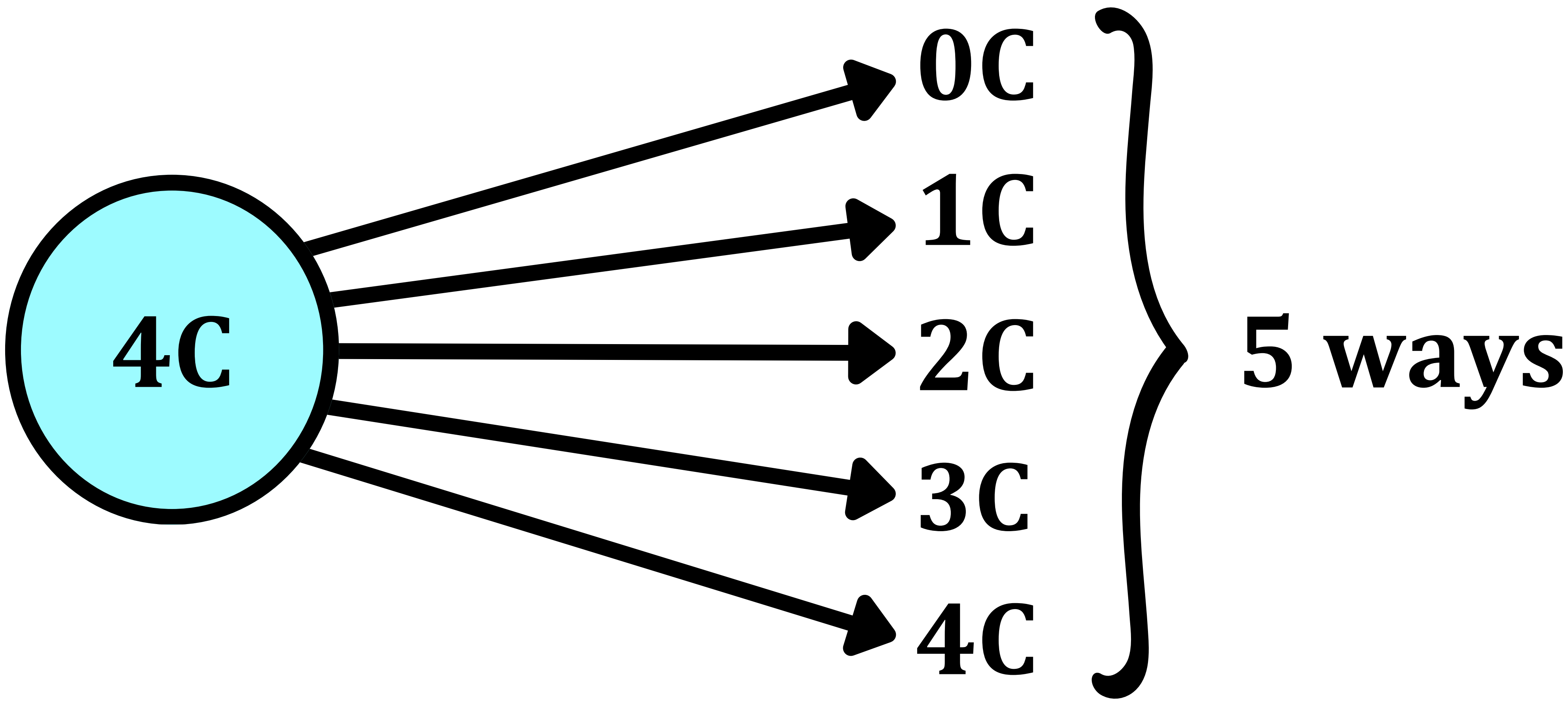


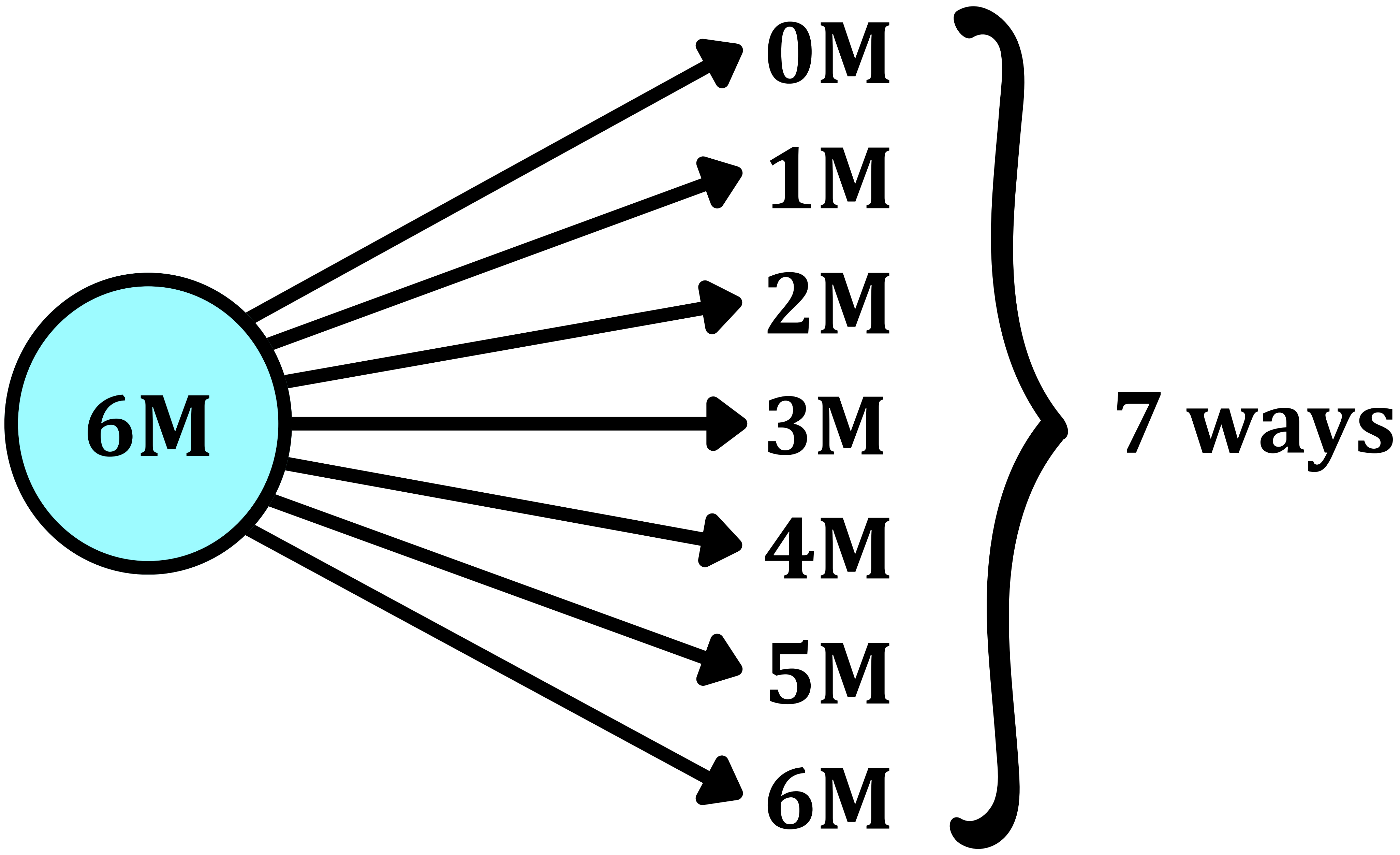
Q12: From a library containing 5 identical physics books, 4 identical chemistry books & 6 identical maths books. Find the number of ways in which we can select:

1. At least 1 book.
2. At least 1 book of Each Subject.
3. At least 2 books.
4. At least 2 books of Each Subject.

1. At least 1 book out of 15 books →







To choose at least 1 book means that it would be okay to choose the 0-book option in any two of the 3 subjects.

But there is a possibility that the 0-book option is chosen for all 3 subjects. To eliminate this possibility, we subtract 1 from the answer.

$$Answer = 6 \times 5 \times 7 - 1 = 209$$

2. At least 1 book of Each Subject →

Therefore, the 0-book option cannot be chosen for any subject.

Physics → 5 ways

Chemistry → 4 ways

Math → 6 ways

$$Answer = 5 \times 4 \times 6 = 120$$

3. At least 2 books out of 15 books →

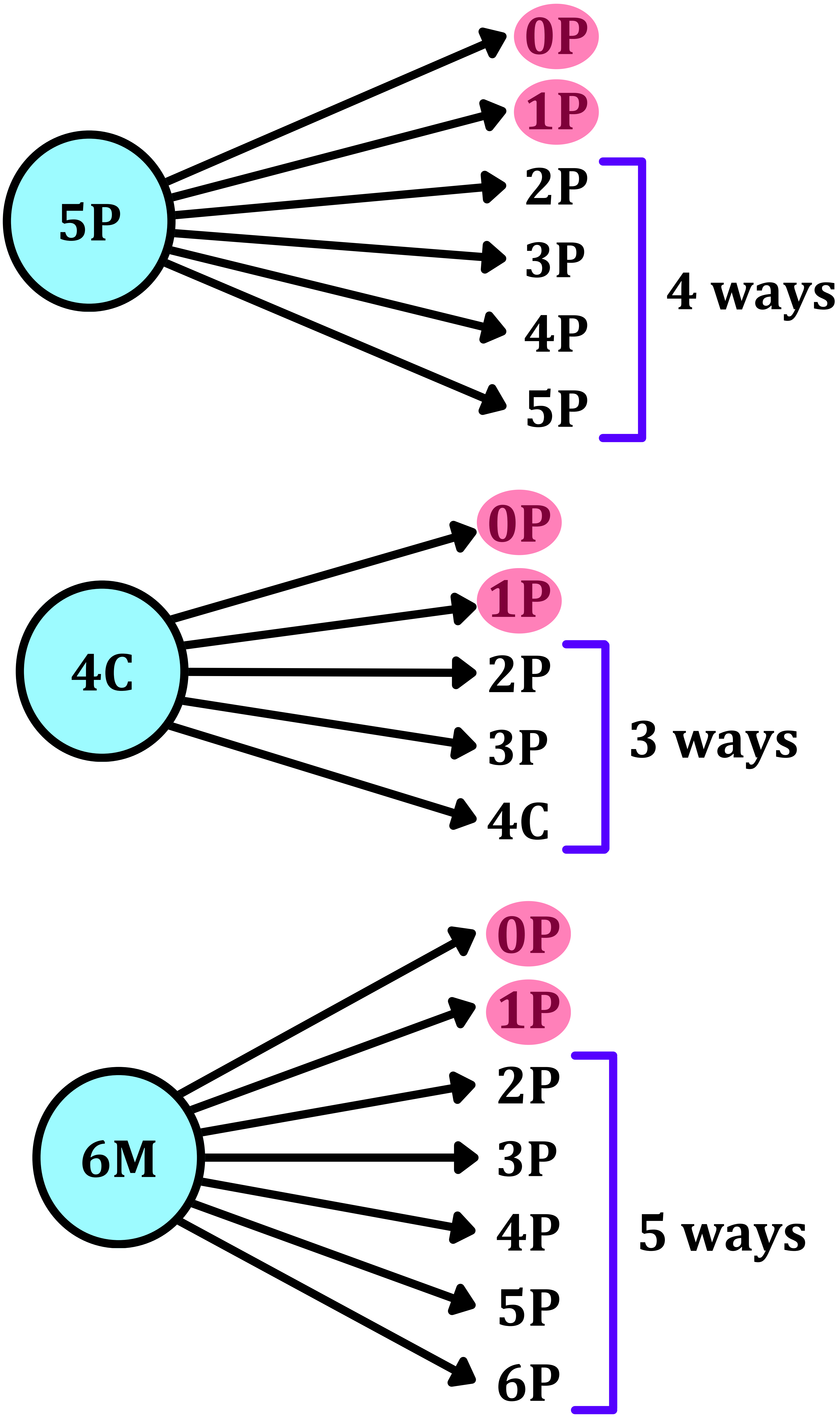
Therefore, we can choose the 0-book option in one of the three subjects provided we don't choose the same in the other two subjects.

0-book option cannot be chosen for all 3 subjects → (–1) possibility.

0-book option cannot be chosen for two subjects simultaneously with the 1-book option for the third one → (–3) possibilities.

$$Answer = 6 \times 5 \times 7 - 1 - 3 = 206$$

4. At least 2 books of Each Subject →



$$Answer = 4 \times 3 \times 5 = 60$$

Q13: From a library containing 5 different physics books, 4 different chemistry books & 6 identical maths books. Find the number of ways in which we can select:

1. At least 1 book.
2. At least 1 book of Each Subject.
3. At least 2 books.
4. At least 2 books of Each Subject.

1. At least 1 book out of 15 books →

Let the 5 Physics Books be $P_1\ P_2\ P_3\ P_4\ P_5$
Let the 4 Chemistry Books be $C_1\ C_2\ C_3\ C_4$
Let the 6 Math Books be M M M M M M

Different ways of selecting:

- **Physics books (different) = 2^5**
- **Chemistry books (different) = 2^4**
- **Math books (same) = n + 1 = 7**

The possibility in which the 0-book option is chosen for all 3 subjects must be eliminated by subtracting 1.

$$Answer = 2^5 . 2^4 . 7 - 1$$

2. At least 1 book of Each Subject →

$$Answer = \left(2^5 - 1\right) \times \left(2^4 - 1\right) \times 6$$

3. At least 2 books out of 15 books →

The possibility of taking 0 books in each subject is eliminated by subtracting 1.

There are 10 possibilities that only one book is selected. These possibilities must be eliminated as well.

- **Physics books:** Since the 5 physics books are different, there are 5 ways to choose exactly one physics book.
- **Chemistry books:** Since the 4 chemistry books are different, there are 4 ways to choose exactly one chemistry book.
- **Maths books**: Since the 6 maths books are identical, there is only 1 way to choose exactly one maths book (as the chosen book is indistinguishable from any other single maths book)

$$Answer = 2^5 . 2^4 .7 - 1 - 10$$

3. At least 2 books of Each Subject →

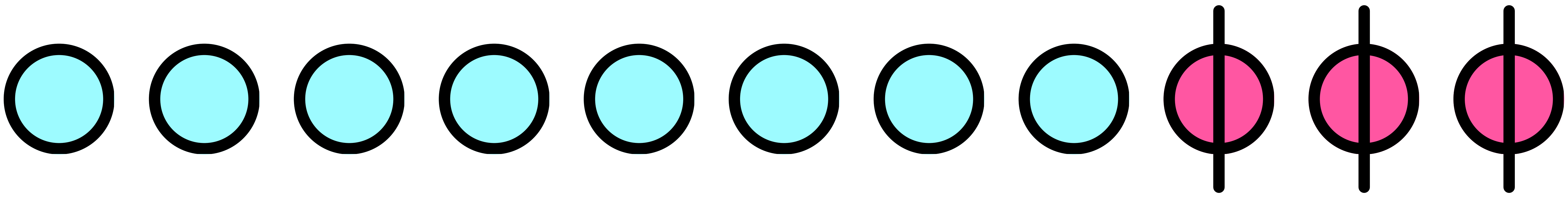
$$(2^5 - 1 - {}^5\mathrm{C}_1)(2^4 - 1 - {}^4\mathrm{C}_1)(2^6 - 1 - {}^6\mathrm{C}_1)$$
$$= (2^5 - 6)(2^4 - 5)(7 - 2)$$
$$= (2^5 - 6)(2^4 - 5)(5)$$

Distribution of Identical Objects (Beggar's Method)

In how many ways can we distribute 8 identical coins to 4 beggars such that each beggar can receive any number of coins?

TRICK →
Add fake coins. The number of fake coins should be one less than the number of beggars.

Arrange these fake coins along with the real coins to get the number of ways in which the distribution can be done.



8 coins → Real

3 coins → Fake

11 coins → Total

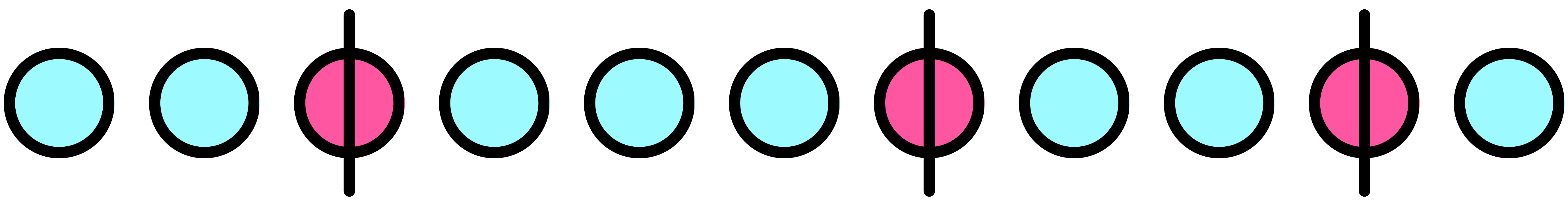
Total Number of Distribution Ways →

$$^{11}C_3 = \frac{11!}{8!\ 3!}$$

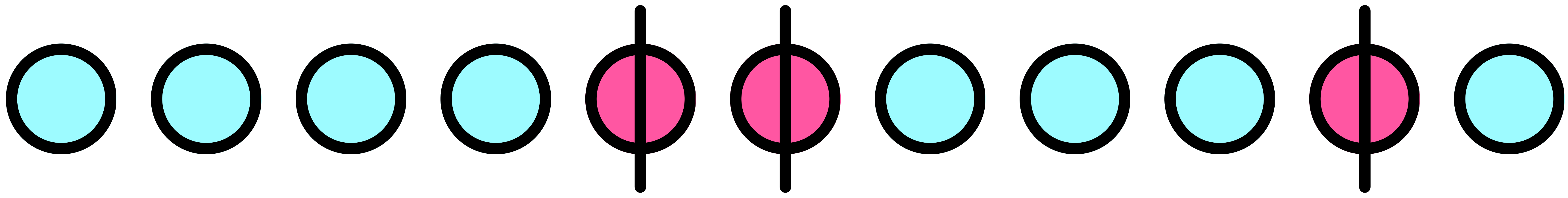
LOGIC →

Each arrangement of these fake and real coins corresponds to one unique distribution pattern.

The fake coins act as partition between the number of coins received by each beggar.



Distribution Pattern → (2,3,2,1)

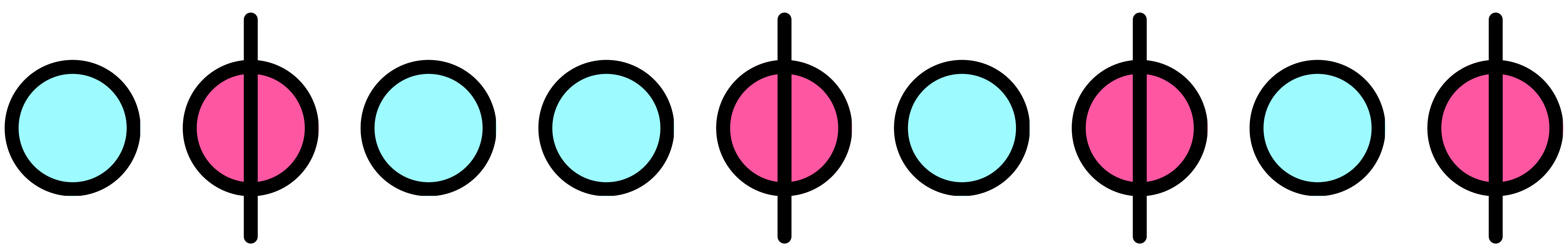


Distribution Pattern → (4,0,3,1)

In how many ways can we distribute 10 identical coins to 5 beggars such that each beggar receives at least one coin?

Since each of the 5 beggars should have at least one coin, we remove 5 coins from the 10 that we have.

Total coins → 10
Coins removed → 5
Fake coins added → 5 – 1 = 4
Coins left → 10 – 5 + 4 = 9



Distribution Pattern (Apart from the mandatory 1 coin per Beggar) → (1,2,1,1,0)

Total Number of Distribution Ways →

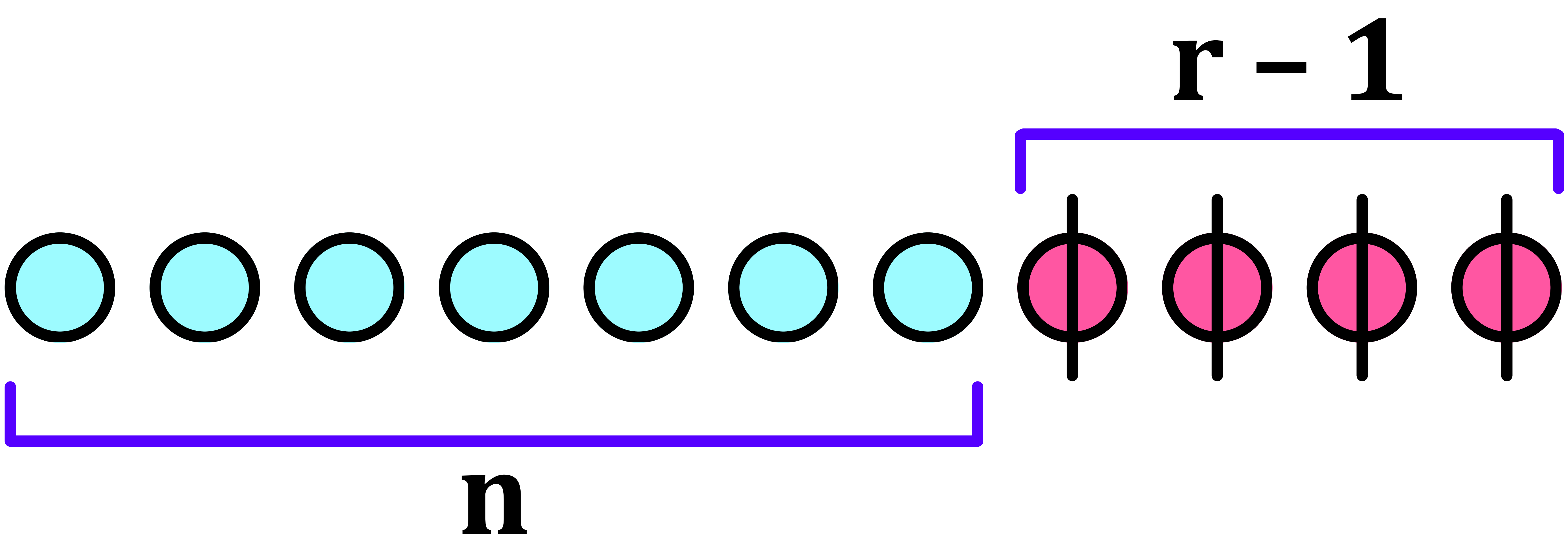
$$^{9}C_{5} = \frac{9!}{5!\ 4!}$$

Converting Beggar's Method into Theorems

THEOREM 1 →

Number of ways to distribute "n" identical objects among "r" persons so that each may get any number of things is give by →

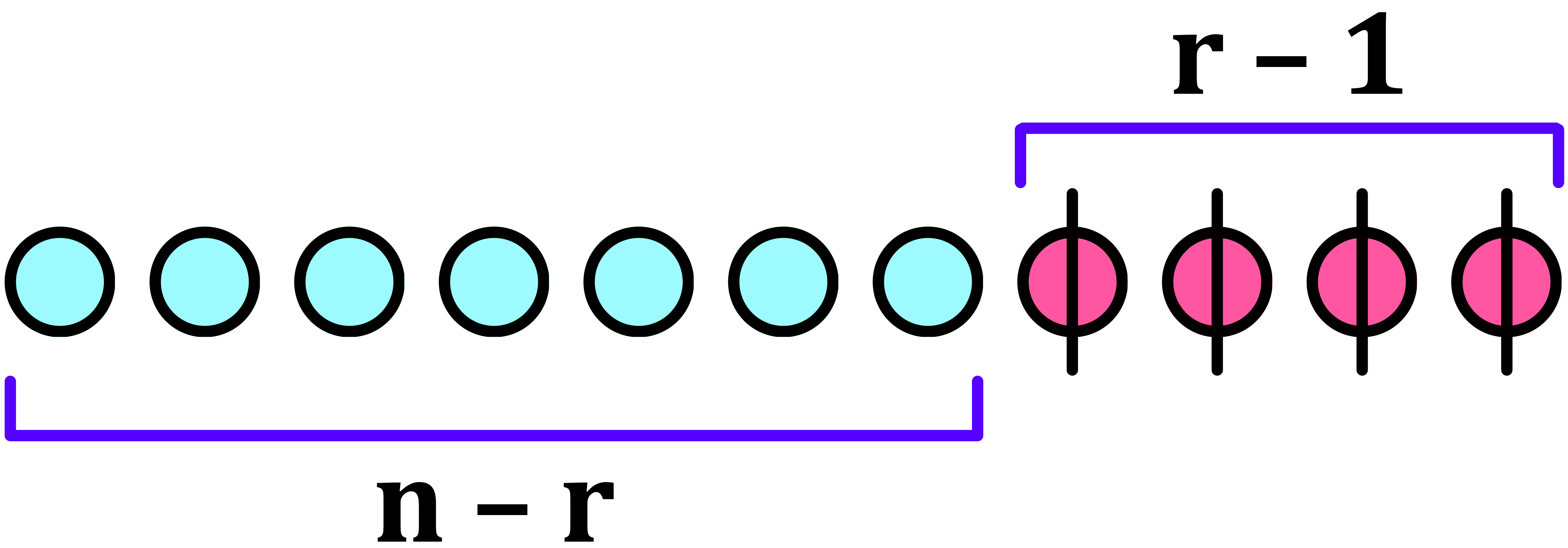
$$^{n+r-1}C_{r-1} = \frac{(n+r-1)!}{n!\,(r-1)!}$$



THEOREM 2 →

Number of ways to distribute "n" identical objects among "r" persons so that each gets at least 1 thing is given by →

$$^{n-1}C_{r-1} = \frac{(n-1)!}{(n-r)!\,(r-1)!}$$



Finding Number of Non Negative Integral Solutions

$$x_1 + x_2 + x_3 + x_4 + x_5 = 20$$

TRICK →

Use the Beggar's Method by considering the variables (x_1 x_2 x_3 x_4 x_5) beggars and the numerical value (20) the total number of coins.

No. of Beggars ⟶ 5

No. of Real Coins ⟶ 20

No. of Fake Coins ⟶ 5 – 1 = 4

Total Coins (Fake + Real) ⟶ 24

$$^{20+4}\mathbf{C}_4 = {}^{24}\mathbf{C}_4$$

$$^{24}\mathbf{C}_4 = \frac{24!}{20!\ 4!}$$

Finding Number of Integral Solutions

CASE 1 →

When each variable has a minimum value:

$$x_1 + x_2 + x_3 + x_4 = 20$$

$$x_1 \geq 0 \ , \ \ x_2 \geq 1 \ , \ \ x_3 \geq 2 \ , \ \ x_4 \geq 4$$

TRICK →

This can be correlated with the second case of the Beggar's Method where every beggar (variable) must get at least 1 coin.
But in this case, the only difference is that the minimum value need not be 1.

Minimum Value (of Coins) Required for Each Variable (Beggar) →

$x_1 \rightarrow 0$

$x_2 \rightarrow 1$

$x_3 \rightarrow 2$

$x_4 \rightarrow 4$

Total Minimum Value (of Coins) Required:

0 + 1 + 2 + 4 = 7

Total Numerical Value (or Total No. of Coins):

20

Subtracting Minimum Value from Total:

20 – 7 = 13

Therefore, only 13 of the 20 coins are useful.

Total no. of Coins = 20

Coins to be Removed = 7

Coins Remaining (Useful Coins) = 13

No. of Beggars (Variables) = 4

Fake Coins Introduced = 4 – 1 = 3

Final Number of Coins (Real + Fake)

= 13 + 3 = 16

ANSWER →

$$^{13+3}\mathbf{C}_3 = {}^{16}\mathbf{C}_3 = \frac{16!}{3!\ 13!}$$

CASE 2 →

When a variable has a minimum negative value:

$$x_1 + x_2 + x_3 = 15$$

$$x_1 \geq 4, \quad x_2 \geq -3, \quad x_3 \geq -1$$

METHOD TO SOLVE →

In case 1, the variables had minimum positive values which were subtracted from the total numerical value.

In case 2, the variables have minimum negative values which should also be subtracted from the total numerical value in a similar manner.

$$15 - (-3) - (-1) - 4 = 15$$

$$15 + 3 + 1 - 4 = 15$$

Total Coins → 15

Total Number of Beggars → 3

Coins Removed → 4 + (– 3) + (– 1) = 0

Fake Coins Added → 3 – 1 = 2

Final Number of Coins (Fake + Real) →
15 + 2 = 17

$$^{15+2}\mathbf{C}_2 = {}^{17}\mathbf{C}_2 = \frac{17!}{2!\ 15!}$$

CASE 3 →

When the entire equation has a maximum value

$$x_1 + x_2 + x_3 + x_4 \le 12, \quad x_i \ge 0$$

TRICKS →

Introduce a fake variable (Fake Beggar), say N, and solve the problem by removing the inequality and replacing it with an equality.

$$x_1 + x_2 + x_3 + x_4 + N = 12$$
$$x_i \ge 0, \quad N \ge 0$$

Total Coins → 12
Total Number of Beggars → 4 + 1 = 5
Fake Coins Added → 5 – 1 = 4
Final Number of Coins → 12 + 4 = 16

$$^{12+4}\mathbf{C}_4 = {}^{16}\mathbf{C}_4 = \frac{16!}{4!\ 12!}$$

CASE 4 →

When the entire equation is lesser than a fixed numerical value

$$x_1 + x_2 + x_3 < 15$$

$$x_1 \geq 0, \quad x_2 \geq 2, \quad x_3 \geq -1$$

Since the questions only demand integral solutions, the question can be rewritten as follows →

$$x_1 + x_2 + x_3 \leqslant 14$$

This can be solved like the previous cases.

Total No. of Coins = 14

Coins to be Removed = 0 + 2 + (– 1) = 1

Coins Remaining (useful coins) = 13

No. of Beggars (variables) = 3

Fake Coins Introduced = 3 – 1 = 2

Final No. of Coins (Useful + Fake) = 15

$$^{12+4}\mathbf{C}_4 = {}^{16}\mathbf{C}_4 = \frac{16!}{4!\ 12!}$$

CASE 5 →

When the entire equation is greater than a fixed numerical value

$$x_1 + x_2 + x_3 \geq 20, \quad x_i \geq 0$$

Such questions are trick questions because they can have infinite solutions.

CASE 5 →

Equation with repeating variables (Identical Beggars)

$$x_1 + x_2 + x_3 + 3x_4 = 20, \quad x_i \geq 0$$

METHOD TO SOLVE →

Divide the question into several cases, each with a different value of x_4 (the repeating variable). This sounds like a lengthy process but it can be simplified very easily.

Case 1 →

$$X_4 = 0 \quad X_1 + X_2 + X_3 = 20$$

Solution:

$$^{22}\mathbf{C}_2$$

Case 2 →

$$X_4 = 1 \quad X_1 + X_2 + X_3 = 17$$

Solution:

$$^{19}\mathbf{C}_2$$

Case 3 →

$$X_4 = 2 \quad X_1 + X_2 + X_3 = 14$$

Solution:

$$^{16}\mathbf{C}_2$$

As you can see, a pattern seems to unfold:

$$ {}^{22}\mathbf{C}_2 \xleftrightarrow{-3} {}^{19}\mathbf{C}_2 \xleftrightarrow{-3} {}^{16}\mathbf{C}_2 $$

Case 1 **Case 2** **Case 3**

Continue following this pattern →

$$ {}^{13}\mathbf{C}_2 \quad {}^{10}\mathbf{C}_2 \quad {}^{7}\mathbf{C}_2 \quad {}^{4}\mathbf{C}_2 $$

Case 4 **Case 5** **Case 6** **Case 7**

Final Case (Case 7) →

$$X_4 = 6 \quad X_1 + X_2 + X_3 = 2$$

Solution:

$$ {}^{4}\mathbf{C}_2 $$

Adding up all of these combinations would give us the answer.